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CLINICAL RESEARCH / ANESTHESIA & PAIN MANAGEMENT

COMPARISON OF IV PARACETAMOL VS KETOROLAC

FOR POSTOPERATIVE PAIN CONTROL IN ELECTIVE HERNIA REPAIR SURGERY UNDER SPINAL ANESTHESIA



RESEARCH REPORT | Statistical Analysis Using jamovi

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Abstract

Intravenous (IV) Paracetamol and IV Ketorolac were compared for postoperative pain management among 118 patients undergoing elective hernia repair surgery under spinal anaesthesia. Each treatment group included 59 patients. We assessed postoperative pain severity using Visual Analogue Scale (VAS) categories. We also examined patient satisfaction, adequacy of pain relief, and adverse effects. The Chi-Square and Mann-Whitney U Tests showed no statistically significant difference in postoperative pain severity between the treatment groups ($p = 0.997$ and $p = 0.914$, respectively). Patient satisfaction ($p < 0.001$) and adequacy of pain relief ($p = 0.028$) differed significantly, with higher satisfaction and adequate pain relief reported among patients receiving IV Paracetamol. Adverse effects did not differ significantly between the groups ($p = 0.261$). IV Paracetamol was associated with more favourable patient-reported outcomes, although measured postoperative pain severity was similar between treatments.

Table of Content

Contents

Abstract 2

Table of Content 3

List of Figures 6

List of Tables 9

List of Abbreviations 10

Statistical Symbols 11

1. Introduction 12

    1.1 Background of the Study 12

    1.2 Problem Statement 12

    1.3 Research Aim 12

    1.4 Research Objectives 12

    1.5 Research Questions 12

    1.6 Research Hypotheses 12

2. Research Methodology 13

    2.1 Study Design 13

    2.2 Study Setting and Duration 13

    2.3 Study Population 13

    2.4 Sample Size Calculation with Formula 13

    2.5 Sampling Technique 13

    2.6 Inclusion Criteria 13

    2.7 Exclusion Criteria 14

    2.8 Data Collection Tool 14

    2.9 Study Variables 14

    2.10 Ethical Considerations 14

    2.11 Statistical Analysis Plan 14

3. Data Preparation in SPSS 15

    3.1 Original Excel Dataset 15

    3.2 Importing Data into SPSS 16

    3.3 Data View and Variable View 16

    3.4 Checking Missing Values 17

    3.5 Identifying Data Entry Errors 17

    3.6 Data Cleaning and Validation 18

    3.7 Verifying Patient Eligibility 19

4. Data Coding and Formula Application 23
4.1 Coding Demographic Variables 23
4.2 Coding Clinical Variables 23
4.3 Coding Treatment Groups 24
4.4 Coding VAS Pain Severity 24
4.5 Coding Patient Satisfaction 25
4.6 Coding Adequate Pain Relief 25
4.7 Coding Adverse Effects 26
4.8 Applying Compute Variable Formulas 26
4.9 Recode into Different Variables 28
4.10 Verification of Coded Data 28
5. Descriptive Statistical Analysis 29
5.1 Frequencies and Percentages 29
5.2 Age and Gender Distribution 30
5.3 BMI and ASA Status 31
5.4 Distribution of Treatment Groups 32
5.5 VAS Pain Severity 32
5.6 Patient Satisfaction 33
5.7 Adequacy of Pain Relief 34
5.8 Adverse Effects and Their Types 34
6. Inferential Statistical Analysis 35
6.1 Selection and Assumptions of Statistical Tests 35
6.2 Chi-Square Test for VAS Pain Severity 36
6.3 Fisher's Exact Test 36
6.4 Mann-Whitney U Test 37
6.5 Chi-Square Test for Patient Satisfaction 37
6.6 Chi-Square Test for Adequate Pain Relief 38
6.7 Chi-Square Test for Adverse Effects 38
6.8 P-Values and Statistical Significance 39
6.9 Summary of Inferential Results 39
7. Graphical Presentation in SPSS 40
7.1 Pie Chart of Patient Satisfaction 40
7.2 Bar Chart of VAS Pain Severity 40
7.3 Bar Chart of Age Distribution 41
7.4 Bar Chart of BMI Categories 41

7.5 Bar Chart of Adequate Pain Relief 42
7.6 Grouped Bar Chart Comparing Paracetamol and Ketorolac 42
7.7 Bar Chart of Adverse Effects 43
8. Results and Interpretation 43
8.1 Descriptive Statistical Findings 43
8.2 Postoperative Pain Control Comparison 44
8.3 Patient Satisfaction Comparison 44
8.4 Adequacy of Pain Relief Comparison 45
8.5 Adverse Effects Comparison 45
8.6 Interpretation of P-Values 46
8.7 Statistical Results Summary Table 46
9. Discussion 47
9.1 Interpretation of Major Findings 47
9.2 Comparison with Previous Research 47
9.3 Clinical Implications 48
9.4 Limitations of the Study 49
10. Conclusion and Recommendations 50
10.1 Conclusion 50
10.2 Recommendations 50
10.3 Suggestions for Future Research 51
References 51

List of Figures

Figure 1:Original Excel Dataset Containing Patient Records and Study Variables 15
Figure 2:Original Excel Dataset Successfully Imported into IBM SPSS Statistics 16
Figure 3: Patient Records, Variable Properties, and Measurement Levels Displayed in IBM SPSS Statistics Data View and Variable View 16
Figure 4:Assessment of Valid and Missing Values Across 22 Study Variables in IBM SPSS Statistics 17
Figure 5: Identification of Data Entry Errors and Inconsistent Categories Using SPSS Frequency Tables.....17
Figure 6: Frequency Distribution of Cleaned IV Paracetamol and IV Ketorolac Treatment Groups in IBM SPSS Statistics 18
Figure 7:Cross-Validation of Reported Adverse Effects and Symptom Descriptions in IBM SPSS Statistics 18
Figure 8: Frequency Distribution of Cleaned and Standardised Adverse-Effect Descriptions in IBM SPSS Statistics 18
Figure 9:Cross-Tabulation of Patient Age Groups and ASA Physical Status for Eligibility Verification in IBM SPSS Statistics 19
Figure 10:Assessment of Valid and Missing Values for Surgical Procedures, Relevant Medical History, and Chronic Pain History in IBM SPSS Statistics 19
Figure 11: Frequency Distribution of Age/ASA Eligibility, Medical History, and Chronic Pain History Among 118 Patients in IBM SPSS Statistics 18
Figure 12:Frequency Distribution and Cross-Tabulation of Kidney Failure, Bleeding Disorders, and Age/ASA Eligibility in IBM SPSS Statistics.....20
Figure 13:Preliminary Patient Eligibility and Treatment Group Distribution Following Age, ASA, and Medical History Screening in IBM SPSS Statistics 21
Figure 14:Verification of Liver-Related Medical History and Surgical Procedures Among Preliminary Eligible Patients in IBM SPSS Statistics 22
Figure 15:Cross-Tabulation of Preliminary Patient Eligibility and Chronic Pain History in IBM SPSS Statistics 22
Figure 16:Frequency Distribution of Numerically Coded Age Groups and Gender in IBM SPSS Statistics 23
Figure 17:Frequency Distribution of Numerically Coded BMI Categories and ASA Physical Status in IBM SPSS Statistics 23
Figure 18:Frequency Distribution and Cross-Validation of Numerically Coded IV Treatment Groups in IBM SPSS Statistics 24
Figure 19:Cross-Validation of Numerically Coded Postoperative VAS Pain Severity Categories in IBM SPSS Statistics 24

Figure 20:Cross-Validation of Numerically Coded Patient Satisfaction Levels in IBM SPSS Statistics 25
Figure 21:Cross-Validation of Numerically Coded Postoperative Pain Relief Adequacy in IBM SPSS Statistics 25
Figure 22: Numeric Coding and Cross-Validation of Postoperative Adverse-Effect Responses in IBM SPSS Statistics 26
Figure 23: Frequency Distribution of Computed Treatment, Postoperative Pain Severity, and Patient Satisfaction Indicators in IBM SPSS Statistics 26
Figure 24: Cross-Validation of Computed Treatment, High Pain, and Positive Satisfaction Indicators Using IBM SPSS Statistics 27
Figure 25: Frequency Distribution and Cross-Validation of Recoded Patient Satisfaction and Postoperative VAS Pain Severity in IBM SPSS Statistics 28
Figure 26: Final Verification of Numeric Codes, Computed Formulas, and Recoded Variables in IBM SPSS Statistics. 28
Figure 27: Frequency and Percentage Distribution of Patient Demographics, Clinical Characteristics, Treatment Groups, and Postoperative Outcomes in IBM SPSS Statistics. 29
Figure 28: Frequency Distribution, Bar Charts, and Crosstabulation of Patient Age Groups and Gender in IBM SPSS Statistics (N = 118). 30
Figure 29: Frequency Distribution, Bar Charts, and Crosstabulation of Patient BMI Categories and ASA Physical Status in IBM SPSS Statistics (N = 118). 31
Figure 30: Frequency Distribution, Bar Chart, and Pie Chart of IV Paracetamol and IV Ketorolac Treatment Groups in IBM SPSS Statistics (N = 118). 32
Figure 31: Frequency Distributions, Bar Chart, Pie Chart, Crosstabulation, and Treatment Group Comparison of Postoperative VAS Pain Severity in IBM SPSS Statistics (N = 118). 32
Figure 32: Grouped Bar Chart Comparing Patient Satisfaction Levels Between IV Paracetamol and IV Ketorolac Groups (N = 118). 33
Figure 33: Frequency Distribution, Bar Chart, Pie Chart, Crosstabulation, and Treatment Group Comparison of Adequacy of Postoperative Pain Relief (N = 118). 34
Figure 34: Frequency Tables, Bar Chart, Pie Chart, Treatment Group Crosstabulation, Grouped Bar Chart, and Distribution of Recorded Postoperative Adverse Effect Types Among Patients Receiving IV Paracetamol and IV Ketorolac (N = 118). 34
Figure 35: Verification of Treatment Group Distribution and Expected Cell Frequencies for Postoperative Pain Severity, Patient Satisfaction, Adequacy of Pain Relief, and Adverse Effects in IBM SPSS Statistics (N = 118). 35
Figure 36: Pearson Chi-Square Test Results for Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118). 36

Figure 37: Fisher-Freeman-Halton Exact Test for Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118). 36
Figure 38: Mann-Whitney U Test Comparing Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118). 37
Figure 39: Pearson Chi-Square and Fisher-Freeman-Halton Exact Test Results for Patient Satisfaction Between IV Paracetamol and IV Ketorolac Groups (N = 118). 37
Figure 40: Pearson Chi-Square Test Comparing Adequacy of Postoperative Pain Relief Between IV Paracetamol and IV Ketorolac Groups (N = 118). 38
Figure 41: Pearson Chi-Square Test Results for Postoperative Adverse Effects Between IV Paracetamol and IV Ketorolac Groups (N = 118). 38
Figure 42: Frequency Distribution and Pie Chart of Patient Satisfaction Levels Following Postoperative Pain Management (N = 118). 40
Figure 43: Frequency Distribution and Bar Chart of Postoperative VAS Pain Severity Among 118 Patients. 40
Figure 44: Frequency Distribution and Bar Chart of Patient Age Groups Among 118 Patients. 41
Figure 45: Frequency Distribution and Bar Chart of Patient BMI Categories Among 118 Patients. 41
Figure 46: Frequency Distribution and Bar Chart of Adequacy of Postoperative Pain Relief Among 118 Patients. 42
Figure 47: Grouped Bar Chart and Crosstabulation Comparing Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118). 42
Figure 48: Frequency Distribution and Bar Chart of Postoperative Adverse Effects Among 118 Patients. 43

List of Tables

Table 1: Statistical Significance of Postoperative Outcomes Between IV Paracetamol and IV Ketorolac Groups (N = 118). 39
Table 2: Summary of Inferential Statistical Results Comparing IV Paracetamol and IV Ketorolac (N = 118).....46

List of Abbreviations

Abbreviation	Full Form
ARDS	Acute Respiratory Distress Syndrome
ASA	American Society of Anesthesiologists
BMI	Body Mass Index
COX-2	Cyclooxygenase-2
ESRA	European Society of Regional Anaesthesia and Pain Therapy
GI	Gastrointestinal
HCV	Hepatitis C Virus
IV	Intravenous
NSAID	Non-Steroidal Anti-Inflammatory Drug
PROSPECT	Procedure-Specific Postoperative Pain Management
SpO_2	Peripheral Oxygen Saturation
SPSS	Statistical Package for the Social Sciences
VAS	Visual Analogue Scale

Statistical Symbols

Symbol	Meaning
H_0	Null Hypothesis
H_1	Alternative Hypothesis
N	Total Sample Size
n	Sample Size
p	Probability Value
U	Mann-Whitney U Test Statistic
Z	Standardised Test Statistic
χ^2	Chi-Square Test Statistic
%	Percentage

1. Introduction

1.1 Background of the Study

Postoperative pain is a common concern among patients undergoing elective hernia repair surgery. Pain can affect patient comfort, movement, sleep, and recovery. Spinal anaesthesia provides pain relief during surgery, but discomfort may develop when its effects decrease. Proper pain management supports patient recovery and satisfaction.

Intravenous (IV) Paracetamol and IV Ketorolac are commonly used medicines for postoperative pain management. Paracetamol helps reduce pain, while Ketorolac is a non-steroidal anti-inflammatory drug (NSAID) that reduces pain and inflammation. Coppens et al. (2020) recommend Paracetamol and NSAIDs as part of pain management following open inguinal hernia repair.

1.2 Problem Statement

Postoperative pain may continue despite the administration of analgesic medicines. Patients receiving different treatments may report variations in pain severity, satisfaction, adequate pain relief, and adverse effects. Comparing IV Paracetamol and IV Ketorolac can help identify differences in patient-reported outcomes following elective hernia repair under spinal anaesthesia.

1.3 Research Aim

To compare IV Paracetamol and IV Ketorolac for postoperative pain management among patients undergoing elective hernia repair surgery under spinal anaesthesia.

1.4 Research Objectives

1. To assess postoperative pain severity in both treatment groups.
2. To compare patient satisfaction with postoperative pain management.
3. To evaluate the adequacy of pain relief between treatment groups.
4. To compare the occurrence of reported adverse effects.

1.5 Research Questions

1. Does postoperative pain severity differ between IV Paracetamol and IV Ketorolac?
2. Is patient satisfaction different between the treatment groups?
3. Does the adequacy of postoperative pain relief differ between treatments?
4. Is there a difference in reported adverse effects?

1.6 Research Hypotheses

H0 (Null Hypothesis): No statistically significant difference exists in postoperative pain outcomes between patients receiving IV Paracetamol and IV Ketorolac.

H1 (Alternative Hypothesis): A statistically significant difference exists in at least one postoperative pain outcome between patients receiving IV Paracetamol and IV Ketorolac.

2. Research Methodology

2.1 Study Design

A comparative cross-sectional study design was selected to examine postoperative pain management among patients receiving intravenous (IV) Paracetamol or IV Ketorolac following elective hernia repair surgery under spinal anaesthesia. Pain severity, patient satisfaction, adequacy of pain relief, and adverse effects were assessed using recorded patient responses. The design allowed comparison of the two treatment groups without assuming random treatment allocation.

2.2 Study Setting and Duration

The approved research synopsis identified Children's Hospital Lahore and Social Security Hospital Lahore as the proposed clinical settings. The study planned to run for four months following synopsis approval. The available Excel dataset contains responses recorded between 14 and 16 October 2026. These timestamps do not independently confirm the complete clinical data collection period.

2.3 Study Population

The dataset contained 118 patient records, divided equally between two treatment groups. Group A included 59 patients who received IV Paracetamol, while Group B included 59 patients who received IV Ketorolac. Patient information covered demographic characteristics, clinical conditions, surgical details, and postoperative outcomes.

2.4 Sample Size Calculation with Formula

The sample size was calculated using a two-proportion comparison formula

$$n = \frac{[(Z_{\alpha/2} + Z_{\beta})^2 \times \{P_1(1-P_1) + P_2(1-P_2)\}]}{(P_1 - P_2)^2}$$

$$n = \frac{[(1.96 + 0.84)^2 \times \{(0.90 \times 0.10) + (0.70 \times 0.30)\}]}{(0.90 - 0.70)^2}$$

$$n = \frac{(7.84 \times 0.30)}{0.04}$$

$$n = 58.8 \approx 59$$

n = 58.8 ≈ 59 patients per group.

The calculated total sample size was 118 patients.

2.5 Sampling Technique

The research synopsis specified convenience sampling. Eligible patients were to be selected based on their availability and willingness to participate. The existing dataset was retained for preliminary statistical examination.

2.6 Inclusion Criteria

The approved synopsis specified the following inclusion criteria:

1. Patients undergoing elective hernia repair surgery.
2. Patients aged 18–60 years.
3. Patients classified as ASA physical status I or II.
4. Patients willing to provide informed consent.

2.7 Exclusion Criteria

The proposed protocol excluded patients with known allergies to Paracetamol or ketorolac. Additional exclusion criteria included renal impairment, liver disease, bleeding disorders, previous chronic analgesic or opioid use, and complicated or emergency hernia surgery.

The available dataset includes patients outside the proposed age and ASA criteria. Their eligibility requires verification before final analysis.

2.8 Data Collection Tool

We recorded patient information using a structured questionnaire and entered it into Microsoft Excel. The dataset included 22 original variables covering demographic information, clinical characteristics, treatment administration, postoperative pain assessment, patient satisfaction, and adverse effects.

2.9 Study Variables

The main independent variable was the administered medication, classified as IV Paracetamol or IV Ketorolac. Dependent variables included postoperative Visual Analogue Scale (VAS) pain severity, patient satisfaction, adequate pain relief, and reported adverse effects. We recorded age, gender, BMI, ASA status, and surgical characteristics as additional variables.

2.10 Ethical Considerations

The research protocol required informed consent and protection of patient information. Patient responses were intended for research purposes, with confidentiality maintained during analysis and reporting. Ethical approval and individual consent records require verification against the original study documentation.

2.11 Statistical Analysis Plan

IBM SPSS Statistics was selected for data coding and analysis. Frequencies and percentages will describe categorical variables. Chi-Square and Fisher's Exact Tests will assess associations between treatment groups and categorical outcomes. The Mann-Whitney U Test will compare ordinal pain severity between independent groups. We will assess statistical significance at $p < 0.05$. Results will be presented through frequency tables, statistical test outputs, and charts.

3. Data Preparation in SPSS

3.1 Original Excel Dataset

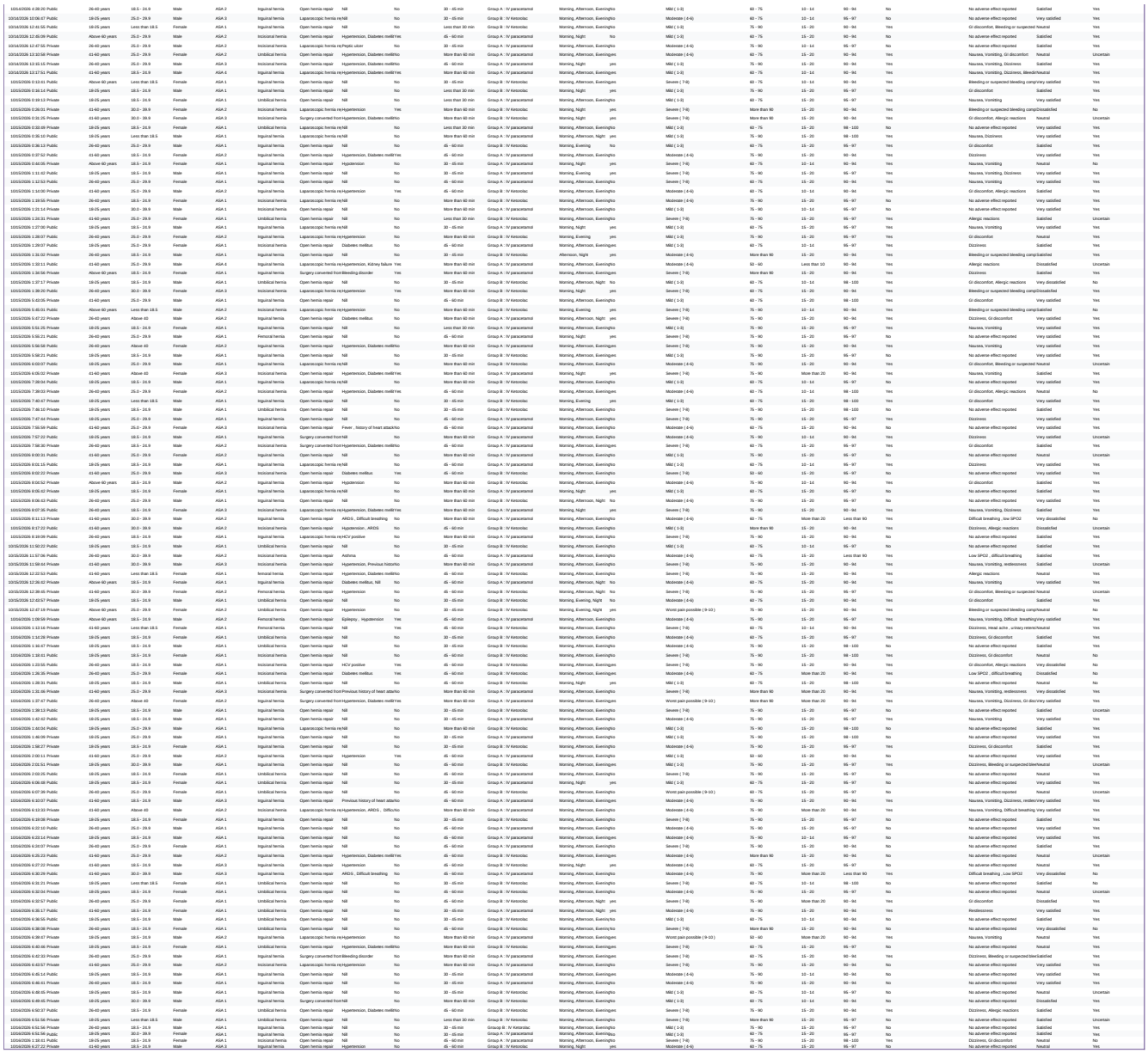


Figure 1: Original Excel Dataset Containing Patient Records and Study Variables

The original dataset was organised in Microsoft Excel before importing it into IBM SPSS Statistics. Figure 1 shows patient records arranged in rows and study variables presented in columns. The dataset includes patient age, gender, BMI, ASA physical status, surgical procedures, treatment groups, postoperative pain severity, patient satisfaction, and adverse effects. We identified 118 patient records and 22 original variables for statistical analysis. The Excel file was retained as the original data source for subsequent data cleaning, coding, and analysis in SPSS.

3.2 Importing Data into SPSS

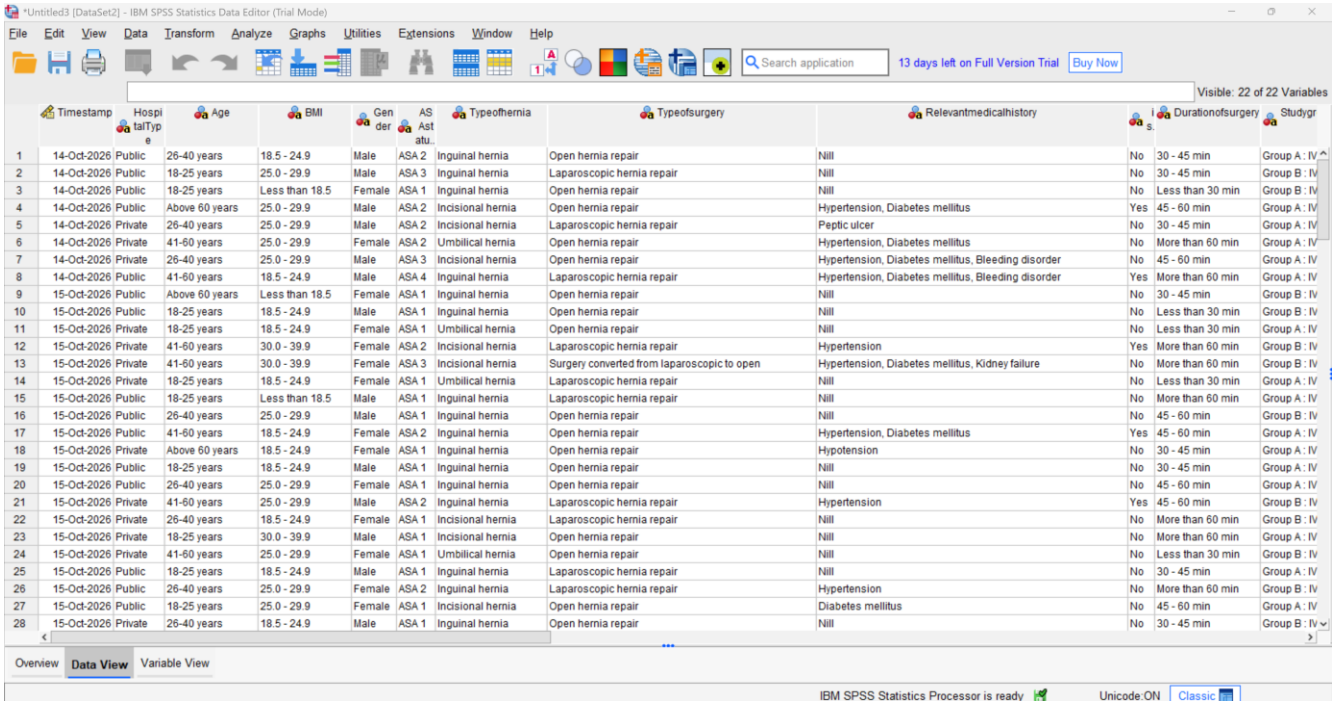


Figure 2:Original Excel Dataset Successfully Imported into IBM SPSS Statistics

The original Excel dataset was imported into IBM SPSS Statistics for data preparation and analysis. Figure 2 shows the imported patient records in SPSS Data View, with each row representing one patient and each column representing a study variable. The dataset contained 118 patient records and 22 original variables, including age, gender, BMI, ASA physical status, type of hernia, surgical procedure, medical history, and treatment group. The imported data were retained for checking variable properties, missing values, and data entry errors before statistical analysis.

3.3 Data View and Variable View

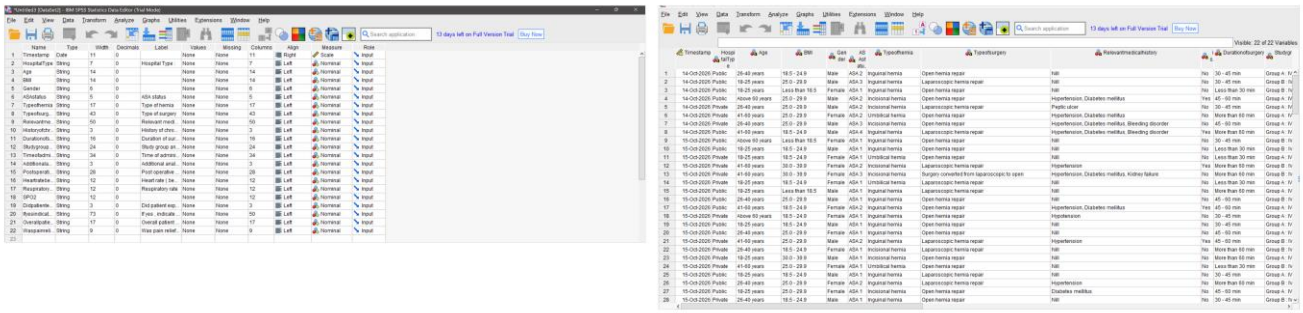


Figure 3: Patient Records, Variable Properties, and Measurement Levels Displayed in IBM SPSS Statistics Data View and Variable View

The original Excel dataset was imported into IBM SPSS Statistics for data preparation and analysis. The Data View displays patient records in rows and study variables in columns, including age, gender, BMI, ASA physical status, type of hernia, surgical procedure, and treatment group. The Variable View presents the properties of 22 imported variables, including variable names, data types, labels, missing value settings, and measurement levels. Most categorical variables were imported as strings with nominal measurement levels. These properties were reviewed to identify variables requiring further coding and adjustment before statistical analysis.

3.4 Checking Missing Values

Statistics

		Timestamp	Hospital Type :	Age	BMI	Gender	ASA status	Type of hernia	Type of surgery	Relevant medical history	History of chronic pain	Duration of surgery	Study group and drug administration	Time of administration	Additional analgesics used	Post operative pain management assessment through VAS	Heart rate (beats / min)	Respiratory rate	SPO2	Did patient experienced any adverse effects	If yes , indicate the adverse effect	Overall patient satisfaction	Was pain relief adequate
N	Valid	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118	118
	Missing	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 4:Assessment of Valid and Missing Values Across 22 Study Variables in IBM SPSS Statistics

Missing values were checked using the Frequencies function in IBM SPSS Statistics. The analysis included 118 patient records and 22 original variables, covering demographic information, clinical characteristics, treatment groups, postoperative pain severity, patient satisfaction, and adverse effects. The SPSS output showed 118 valid responses and zero missing values for every variable. No missing values were detected under the current SPSS settings. Text responses such as "Nil" and entries relating to adverse-effect types require further checking to confirm that they were recorded correctly. These checks will be completed during data cleaning and validation.

3.5 Identifying Data Entry Errors

BMI

Category	Frequency	%
18.5 - 24.9	57	48.3%
25.0 - 29.9	34	28.8%
30.0 - 39.9	13	11.0%
Above 40	5	4.2%
Less than 18.5	9	7.6%
Total	**118**	**100.0%**

Age

Category	Frequency	%
18-25 years	49	41.5%
26-40 years	30	25.4%
41-60 years	30	25.4%
Above 60 years	9	7.6%
Total	**118**	**100.0%**

Gender

Category	Frequency	%
Female	50	42.4%
Male	68	57.6%
Total	**118**	**100.0%**

ASA Status

Category	Frequency	%
ASA 1	74	62.7%
ASA 2	27	22.9%
ASA 3	15	12.7%
ASA 4	2	1.7%
Total	**118**	**100.0%**

Study Group and Drug Administration

Category	Frequency	%	
Grouop B : IV Ketorolac	1	0.8%	← spelling inconsistency detected
Group A : IV paracetamol	59	50.0%	
Group B : IV Ketorolac	58	49.2%	
Total	**118**	**100.0%**	

Patient Satisfaction

Category	Frequency	%
Very dissatisfied	6	5.1%
Dissatisfied	7	5.9%
Neutral	28	23.7%
Satisfied	37	31.4%
Very satisfied	40	33.9%
Total	**118**	**100.0%**

Adequacy of Pain Relief

Category	Frequency	%
No	16	13.6%
Uncertain	17	14.4%
Yes	85	72.0%
Total	**118**	**100.0%**

VAS Pain Categories

Category	Frequency	%
Mild (1-3)	36	30.5%
Moderate (4-6)	37	31.4%
Severe (7-8)	41	34.7%
Worst pain possible (9-10)	4	3.4%
Total	**118**	**100.0%**

Adverse Effects Reported

Category	Frequency	%
No	48	40.7%
Yes	70	59.3%
Total	**118**	**100.0%**

Adverse Effect Descriptions Reviewed

capitalization, spacing, and spelling inconsistencies present

Description	n
Allergic reactions	3
Bleeding or suspected bleeding complication	4
Bleeding or suspected bleeding complication, Allergic reactions	2
Difficult breathing, low SPO2	1
Difficult breathing, Low SPO2	1
Dizziness	6
Dizziness, Allergic reactions	2
Dizziness, Bleeding or suspected bleeding complication	2
Dizziness, GI discomfort	6
Dizziness, Head ache, urinary retension	1
GI discomfort	9
GI discomfort, Allergic reactions	5
GI discomfort, Bleeding or suspected bleeding complication	2
GI discomfort, Bleeding or suspected bleeding complication, Hypotension	1
Low SPO2, difficult breathing	2
Nausea, Dizziness	1
Nausea, Vomitting	10
Nausea, Vomitting, Difficult breathing	1
Nausea, Vomitting, Dizziness	3
Nausea, Vomitting, Dizziness, Bleeding or suspected bleeding complication	1
Nausea, Vomitting, Dizziness, GI discomfort	1
Nausea, Vomitting, Dizziness, restlessness	1
Nausea, Vomitting, GI discomfort	1
Nausea, Vomitting, restlessness	2
No adverse effect reported	48
Restlessness	1
Total	**118**

Key Data Entry Issues Identified

- Missing values were not detected in the reviewed variables.
- One treatment-group spelling error was found: "Grouop B : IV Ketorolac".
- Adverse-effect entries showed inconsistent capitalization and spacing.
- Spelling inconsistencies included terms such as "Vomitting" and "urinary retension".

Figure 5: Identification of Data Entry Errors and Inconsistent Categories Using SPSS Frequency Tables

Data entry errors were examined using frequency tables generated in IBM SPSS Statistics. The analysis included 118 patient records and covered age, gender, BMI, ASA status, treatment groups, postoperative pain severity, patient satisfaction, and adverse effects. All reviewed variables contained 118 valid responses with no missing values detected. One spelling error was identified in the treatment group variable, where "Grouop B" appeared instead of "Group B". Inconsistent spelling, capitalisation, and spacing were also found in adverse-effect descriptions, including "Vomitting", "urinary retention", and variations of "Low SPO2". The gender variable contained 68 male and 50 female patients without duplicate categories. These findings identified the entries requiring correction during data cleaning and validation.

3.6 Data Cleaning and Validation

Cleaned IV Treatment Group

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	IV Paracetamol	59	50.0	50.0	50.0
	IV Ketorolac	59	50.0	50.0	100.0
	Total	118	100.0	100.0	

Figure 6: Frequency Distribution of Cleaned IV Paracetamol and IV Ketorolac Treatment Groups in IBM SPSS Statistics



Figure 7:Cross-Validation of Reported Adverse Effects and Symptom Descriptions in IBM SPSS Statistics

Standardised Adverse Effect Descriptions

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Allergic reactions	3	2.5	2.5	2.5
	Bleeding or suspected bleeding complication	4	3.4	3.4	5.9
	Bleeding or suspected bleeding complication, Allergic reactions	2	1.7	1.7	7.6
	Difficult breathing, Low SpO2	4	3.4	3.4	11.0
	Dizziness	6	5.1	5.1	16.1
	Dizziness, Allergic reactions	2	1.7	1.7	17.8
	Dizziness, Bleeding or suspected bleeding complication	2	1.7	1.7	19.5
	Dizziness, GI discomfort	6	5.1	5.1	24.6
	Dizziness, Headache, urinary retention	1	.8	.8	25.4
	GI discomfort	9	7.6	7.6	33.1
	GI discomfort, Allergic reactions	5	4.2	4.2	37.3
	GI discomfort, Bleeding or suspected bleeding complication	2	1.7	1.7	39.0
	GI discomfort, Bleeding or suspected bleeding complication, Hypotension	1	.8	.8	39.8
	Nausea, Dizziness	1	.8	.8	40.7
	Nausea, Vomiting	10	8.5	8.5	49.2
	Nausea, Vomiting, Difficult breathing	2	1.7	1.7	50.8
	Nausea, Vomiting, Dizziness	3	2.5	2.5	53.4
	Nausea, Vomiting, Dizziness, Bleeding or suspected bleeding complication	1	.8	.8	54.2
	Nausea, Vomiting, Dizziness, GI discomfort	1	.8	.8	55.1
	Nausea, Vomiting, Dizziness, restlessness	1	.8	.8	55.9
	Nausea, Vomiting, GI discomfort	1	.8	.8	56.8
	Nausea, Vomiting, restlessness	2	1.7	1.7	58.5
	No adverse effect reported	48	40.7	40.7	99.2
	Restlessness	1	.8	.8	100.0
	Total	118	100.0	100.0	

Figure 8: Frequency Distribution of Cleaned and Standardised Adverse-Effect Descriptions in IBM SPSS Statistics

Data cleaning and validation were performed in IBM SPSS Statistics to improve the consistency and accuracy of the dataset. The treatment group variable was cleaned using the Recode into Different Variables function to correct inconsistent spelling and assign numeric codes. The frequency analysis confirmed that 59 patients (50.0%) received IV Paracetamol and 59 patients (50.0%) received IV Ketorolac, with no missing treatment group values.

Adverse-effect responses were validated using Crosstabs. The results confirmed that 48 patients (40.7%) reported no adverse effects, while 70 patients (59.3%) reported one or more symptoms. All negative responses matched the "No adverse effect reported" category, and all positive responses matched recorded symptoms.

SPSS syntax was used to standardise adverse-effect descriptions by correcting spelling, capitalisation, spacing, and inconsistent symptom combinations. A separate cleaned variable was created to preserve the original responses. The final frequency analysis confirmed 118 valid records with no missing values. These procedures improved data consistency and prepared the dataset for subsequent statistical analysis.

3.7 Verifying Patient Eligibility

Age * ASA status Crosstabulation

Count

		ASA status				
		ASA 1	ASA 2	ASA 3	ASA 4	Total
Age	18-25 years	47	1	1	0	49
	26-40 years	18	9	3	0	30
	41-60 years	5	12	11	2	30
	Above 60 years	4	5	0	0	9
Total		74	27	15	2	118

Figure 9:Cross-Tabulation of Patient Age Groups and ASA Physical Status for Eligibility Verification in IBM SPSS Statistics

The SPSS analysis included 118 patients. Among them, 92 patients (78.0%) were aged 18–60 years and classified as ASA I or II, meeting both eligibility criteria. Nine patients (7.6%) were above 60 years, while 17 patients (14.4%) had ASA III or IV status. These 26 patients (22.0%) did not meet the specified age or ASA requirements. The findings identified differences between the recorded patient characteristics and the approved inclusion criteria.

Frequencies

Statistics

		Type of surgery	Relevant medical history	History of chronic pain
N	Valid	118	118	118
	Missing	0	0	0

Figure 10:Assessment of Valid and Missing Values for Surgical Procedures, Relevant Medical History, and Chronic Pain History in IBM SPSS Statistics

The SPSS frequency analysis included 118 patient records for each of the three variables: type of surgery, relevant medical history, and history of chronic pain. All three variables showed 118 valid responses (100%) and zero missing values. These results confirmed that the selected variables contained no missing entries under the current SPSS settings and were available for further patient eligibility screening.



SPSS Frequency Tables

Patient eligibility screening | 118 records

Type of Surgery

Type of surgery

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Laparoscopic hernia repair	23	19.5	19.5	19.5
	Open hernia repair	87	73.7	73.7	93.2
	Surgery converted from laparoscopic to open	8	6.8	6.8	100.0
	Total	118	100.0	100.0	

History of Chronic Pain

History of chronic pain

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	99	83.9	83.9	83.9
	Yes	19	16.1	16.1	100.0
	Total	119	100.0	100.0	

Relevant Medical History

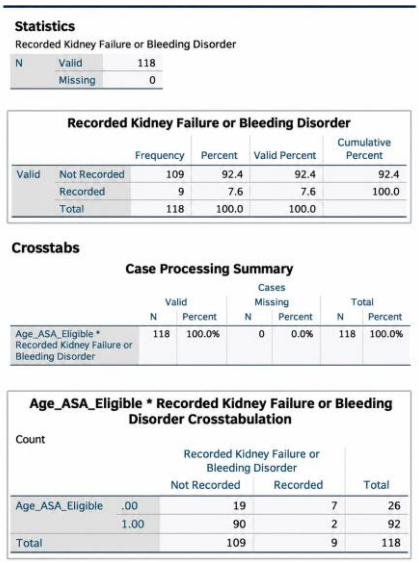
Relevant medical history

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	ARDS , Difficult breathing	2	1.7	1.7	1.7
	Asthma	1	.8	.8	2.5
	Bleeding disorder	2	1.7	1.7	4.2
	Diabetes mellitus	4	3.4	3.4	7.6
	Diabetes mellitus, Nill	1	.8	.8	8.5
	Epilepsy , Hypotension	1	.8	.8	9.3
	Fever , history of heart attack	1	.8	.8	10.2
	HCV positive	2	1.7	1.7	11.9
	Hypertension	13	11.0	11.0	22.9
	Hypertension, ARDS , Difficult breathing	1	.8	.8	23.7
	Hypertension, Diabetes mellitus	10	8.5	8.5	32.2
	Hypertension, Diabetes mellitus, Bleeding disorder	3	2.5	2.5	34.7
	Hypertension, Diabetes mellitus, Kidney failure	3	2.5	2.5	37.3
	Hypertension, Kidney failure	1	.8	.8	38.1
	Hypertension, Previous history of hearth attack	1	.8	.8	39.0
	Hypotension	2	1.7	1.7	40.7
	Hypotension , ARDS	1	.8	.8	41.5
	Nill	66	55.9	55.9	97.5
	Peptic ulcer	1	.8	.8	98.3
	Previous history of heart attack	2	1.7	1.7	100.0
	Total	118	100.0	100.0	

Source: IBM SPSS Statistics frequency output. Original values preserved.

Figure 11: Frequency Distribution of Surgical Procedures, Relevant Medical History, and Chronic Pain History Among 118 Patients in IBM SPSS Statistics

The SPSS analysis included 118 patients. Open hernia repair was the most common procedure, performed in 87 patients (73.7%), followed by laparoscopic repair in 23 patients (19.5%). Eight patients (6.8%) underwent conversion from laparoscopic to open surgery. Medical history records showed that 66 patients (55.9%) reported no relevant medical conditions. Bleeding disorders were recorded in five patients, kidney failure in four, and HCV-positive status in two. Chronic pain history was reported by 19 patients (16.1%), while 99 patients (83.9%) reported no chronic pain. These findings helped identify patients requiring further eligibility verification.



Statistics

Recorded Kidney Failure or Bleeding Disorder

N	Valid	118
	Missing	0

Recorded Kidney Failure or Bleeding Disorder

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not Recorded	109	92.4	92.4	92.4
	Recorded	9	7.6	7.6	100.0
	Total	118	100.0	100.0	

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Age_ASA_Eligible * Recorded Kidney Failure or Bleeding Disorder	118	100.0%	0	0.0%	118	100.0%

Age_ASA_Eligible * Recorded Kidney Failure or Bleeding Disorder Crosstabulation

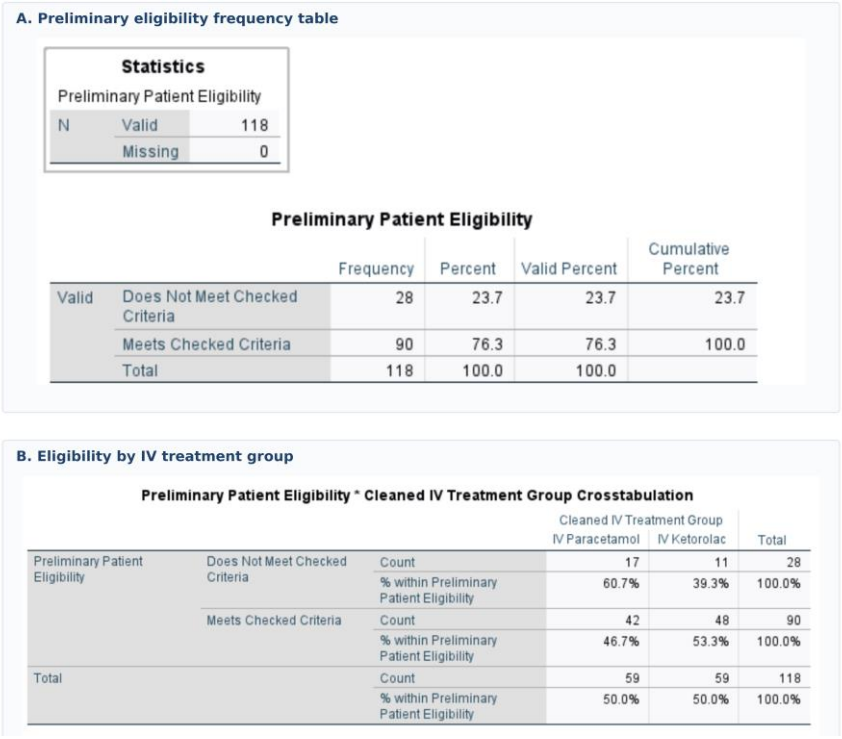
Count

		Recorded Kidney Failure or Bleeding Disorder		
		Not Recorded	Recorded	Total
Age_ASA_Eligible	.00	19	7	26
	1.00	90	2	92
Total		109	9	118

Figure 12:Frequency Distribution and Cross-Tabulation of Kidney Failure, Bleeding Disorders, and Age/ASA Eligibility in IBM SPSS Statistics

The SPSS analysis identified nine patients (7.6%) with recorded kidney failure or bleeding disorders, while 109 patients (92.4%) had neither condition recorded. Among the 92 patients meeting the age

and ASA inclusion criteria, two had these medical conditions, while 90 had no recorded kidney failure or bleeding disorder. Of the 26 patients who did not meet the age and ASA criteria, seven had these conditions. The analysis identified 90 patients (76.3%) who met the checked age, ASA, and medical history criteria. Further clinical verification is required before confirming their final eligibility.



A. Preliminary eligibility frequency table

Statistics

Preliminary Patient Eligibility

N	Valid	118
	Missing	0

Preliminary Patient Eligibility

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Does Not Meet Checked Criteria	28	23.7	23.7	23.7
	Meets Checked Criteria	90	76.3	76.3	100.0
	Total	118	100.0	100.0	

B. Eligibility by IV treatment group

Preliminary Patient Eligibility * Cleaned IV Treatment Group Crosstabulation

			Cleaned IV Treatment Group		
			IV Paracetamol	IV Ketorolac	Total
Preliminary Patient Eligibility	Does Not Meet Checked Criteria	Count	17	11	28
		% within Preliminary Patient Eligibility	60.7%	39.3%	100.0%
	Meets Checked Criteria	Count	42	48	90
		% within Preliminary Patient Eligibility	46.7%	53.3%	100.0%
Total		Count	59	59	118
		% within Preliminary Patient Eligibility	50.0%	50.0%	100.0%

Figure 13:Preliminary Patient Eligibility and Treatment Group Distribution Following Age, ASA, and Medical History Screening in IBM SPSS Statistics

Patient eligibility was assessed using three criteria: age between 18 and 60 years, ASA physical status I or II, and no recorded kidney failure or bleeding disorder. Among 118 patients, 92 met the age and ASA requirements. Two of these patients had recorded medical exclusion conditions, leaving 90 patients (76.3%) who met all three checked criteria. The remaining 28 patients (23.7%) did not meet at least one criterion. Among the 90 preliminary eligible patients, 42 (46.7%) received IV Paracetamol and 48 (53.3%) received IV Ketorolac. The eligibility calculation was $92 - 2 = 90$ patients. Further verification of medication allergies, previous analgesic use, informed consent, and other exclusion criteria is required before confirming the final eligible sample.

Liver-Related Medical History Requiring Review

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not Mentioned	116	98.3	98.3	98.3
	Requires Review	2	1.7	1.7	100.0
	Total	118	100.0	100.0	

Preliminary Patient Eligibility * Liver-Related Medical History Requiring Review Crosstabulation

Count

		Liver-Related Medical History Requiring Review		
		Not Mentioned	Requires Review	Total
Preliminary Patient Eligibility	Does Not Meet Checked Criteria	28	0	28
	Meets Checked Criteria	88	2	90
Total		116	2	118

Preliminary Patient Eligibility * Type of surgery Crosstabulation

Count

		Type of surgery			
		Laparoscopic hernia repair	Open hernia repair	Surgery converted from laparoscopic to open	Total
Preliminary Patient Eligibility	Does Not Meet Checked Criteria	6	17	5	28
	Meets Checked Criteria	17	70	3	90
Total		23	87	8	118

Figure 14:Verification of Liver-Related Medical History and Surgical Procedures Among Preliminary Eligible Patients in IBM SPSS Statistics

The SPSS analysis identified two patients (1.7%) with liver-related medical history requiring further review, while 116 patients (98.3%) had no liver-related condition mentioned. Both patients requiring review belonged to the 90 preliminary eligible patients, leaving 88 patients without a recorded liver-related flag. Among the 90 preliminary eligible patients, 70 (77.8%) underwent open hernia repair, 17 (18.9%) underwent laparoscopic repair, and three (3.3%) had surgery converted from laparoscopic to open. These findings helped identify patients requiring further clinical verification. Liver-related history and surgical conversion were not considered automatic reasons for exclusion.

Preliminary Patient Eligibility * History of chronic pain Crosstabulation

			History of chronic pain		
			No	Yes	Total
Preliminary Patient Eligibility	Does Not Meet Checked Criteria	Count	18	10	28
		% within Preliminary Patient Eligibility	64.3%	35.7%	100.0%
	Meets Checked Criteria	Count	81	9	90
		% within Preliminary Patient Eligibility	90.0%	10.0%	100.0%
Total		Count	99	19	118
		% within Preliminary Patient Eligibility	83.9%	16.1%	100.0%

Figure 15:Cross-Tabulation of Preliminary Patient Eligibility and Chronic Pain History in IBM SPSS Statistics

Chronic pain history was examined among 118 patients using SPSS Crosstabs. Of the 90 patients meeting the preliminary eligibility criteria, 81 (90.0%) reported no chronic pain history, while 9 (10.0%) reported chronic pain. Among the 28 patients who did not meet the checked criteria, 18 had no chronic pain and 10 reported chronic pain. The preliminary eligible sample remained 90 patients, calculated by subtracting two patients with recorded kidney failure or bleeding disorders from the 92 patients meeting the age and ASA requirements ($92 - 2 = 90$). Chronic pain history alone was not considered an exclusion criterion because it does not confirm regular analgesic or opioid use. Further medication history verification is required before confirming final patient eligibility.

4. Data Coding and Formula Application

4.1 Coding Demographic Variables

Patient Age Group

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-25 years	49	41.5	41.5	41.5
	26-40 years	30	25.4	25.4	66.9
	41-60 years	30	25.4	25.4	92.4
	Above 60 years	9	7.6	7.6	100.0
	Total	118	100.0	100.0	

Patient Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	50	42.4	42.4	42.4
	Male	68	57.6	57.6	100.0
	Total	118	100.0	100.0	

Figure 16:Frequency Distribution of Numerically Coded Age Groups and Gender in IBM SPSS Statistics

Age and gender variables were converted into numeric categories using SPSS syntax. The analysis included 118 patients, with 49 (41.5%) aged 18–25 years, 30 (25.4%) aged 26–40 years, 30 (25.4%) aged 41–60 years, and 9 (7.6%) above 60 years. Gender distribution showed 68 male patients (57.6%) and 50 female patients (42.4%). The coded variables retained all 118 valid responses and were prepared for further statistical analysis.

4.2 Coding Clinical Variables

Patient BMI Category

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Less than 18.5	9	7.6	7.6	7.6
	18.5 - 24.9	57	48.3	48.3	55.9
	25.0 - 29.9	34	28.8	28.8	84.7
	30.0 - 39.9	13	11.0	11.0	95.8
	Above 40	5	4.2	4.2	100.0
	Total	118	100.0	100.0	

Patient ASA Physical Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	ASA 1	74	62.7	62.7	62.7
	ASA 2	27	22.9	22.9	85.6
	ASA 3	15	12.7	12.7	98.3
	ASA 4	2	1.7	1.7	100.0
	Total	118	100.0	100.0	

Figure 17:Frequency Distribution of Numerically Coded BMI Categories and ASA Physical Status in IBM SPSS Statistics

BMI and ASA physical status were converted into numeric categories using SPSS syntax. Among 118 patients, 57 (48.3%) had a BMI of 18.5–24.9, followed by 34 (28.8%) in the 25.0–29.9 category. Nine patients (7.6%) had a BMI below 18.5, while 18 (15.3%) were in categories of 30.0 or above. ASA classification showed 74 patients (62.7%) in ASA I, 27 (22.9%) in ASA II, 15 (12.7%) in ASA III, and two (1.7%) in ASA IV. All 118 records were successfully coded without missing values. The numeric variables were prepared for further statistical analysis.

4.3 Coding Treatment Groups

Cleaned IV Treatment Group

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	IV Paracetamol	59	50.0	50.0	50.0
	IV Ketorolac	59	50.0	50.0	100.0
	Total	118	100.0	100.0	

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Study group and drug administration * Cleaned IV Treatment Group	118	100.0%	0	0.0%	118	100.0%

Study group and drug administration * Cleaned IV Treatment Group Crosstabulation

Count

		Cleaned IV Treatment Group		
		IV Paracetamol	IV Ketorolac	Total
Study group and drug administration	Grouop B : IV Ketorolac	0	1	1
	Group A : IV paracetamol	59	0	59
	Group B : IV Ketorolac	0	58	58
Total		59	59	118

Figure 18:Frequency Distribution and Cross-Validation of Numerically Coded IV Treatment Groups in IBM SPSS Statistics

Treatment groups were converted into numeric codes using IBM SPSS Statistics, with code 1 assigned to IV Paracetamol and code 2 to IV Ketorolac. The original treatment entries were checked against the coded variable using Crosstabs. One incorrectly spelled Ketorolac entry was successfully assigned to the correct treatment group. The frequency analysis confirmed 59 patients (50.0%) in each treatment group, with 118 valid records and no missing values. This coding allowed the treatment groups to be used in subsequent statistical comparisons.

4.4 Coding VAS Pain Severity

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Post operative pain management assessment through VAS * Postoperative VAS Pain Severity	118	100.0%	0	0.0%	118	100.0%

Post operative pain management assessment through VAS * Postoperative VAS Pain Severity Crosstabulation

Count

		Postoperative VAS Pain Severity				
		Mild (1-3)	Moderate (4-6)	Severe (7-8)	Worst Pain (9-10)	Total
Post operative pain management assessment through VAS	Mild (1-3)	36	0	0	0	36
	Moderate (4-6)	0	37	0	0	37
	Severe (7-8)	0	0	41	0	41
	Worst pain possible (9-10)	0	0	0	4	4
Total		36	37	41	4	118

Figure 19:Cross-Validation of Numerically Coded Postoperative VAS Pain Severity Categories in IBM SPSS Statistics

Postoperative pain severity was converted into four numeric categories using SPSS syntax. The analysis showed that 36 patients (30.5%) experienced mild pain, 37 (31.4%) experienced moderate pain, 41 (34.7%) experienced severe pain, and four (3.4%) reported the worst pain category. Severe pain was the most frequently reported category. The cross-tabulation confirmed that all 118 patient responses were correctly assigned to their corresponding numeric codes, with no missing values or coding mismatches. The coded variable was prepared for further statistical analysis.

4.5 Coding Patient Satisfaction

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Overall patient satisfaction * Patient Satisfaction Level	118	100.0%	0	0.0%	118	100.0%

Overall patient satisfaction * Patient Satisfaction Level Crosstabulation

Count

		Patient Satisfaction Level					
		Very dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied	Total
Overall patient satisfaction	Dissatisfied	0	7	0	0	0	7
	Neutral	0	0	28	0	0	28
	Satisfied	0	0	0	37	0	37
	Very dissatisfied	6	0	0	0	0	6
	Very satisfied	0	0	0	0	40	40
Total		6	7	28	37	40	118

Figure 20:Cross-Validation of Numerically Coded Patient Satisfaction Levels in IBM SPSS Statistics

Patient satisfaction responses were converted into five numeric categories using SPSS syntax. Among 118 patients, 40 (33.9%) were very satisfied, 37 (31.4%) were satisfied, 28 (23.7%) were neutral, seven (5.9%) were dissatisfied, and six (5.1%) were very dissatisfied. A total of 77 patients (65.3%) reported positive satisfaction levels. The cross-tabulation confirmed that all 118 responses were assigned to the correct numeric codes, with no missing values or coding errors. The coded variable was prepared for further statistical analysis.

4.6 Coding Adequate Pain Relief

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Was pain relief adequate * Adequacy of Postoperative Pain Relief	118	100.0%	0	0.0%	118	100.0%

Was pain relief adequate * Adequacy of Postoperative Pain Relief Crosstabulation

Count

		Adequacy of Postoperative Pain Relief			
		No	Uncertain	Yes	Total
Was pain relief adequate	No	16	0	0	16
	Uncertain	0	17	0	17
	Yes	0	0	85	85
Total		16	17	85	118

Figure 21:Cross-Validation of Numerically Coded Postoperative Pain Relief Adequacy in IBM SPSS Statistics

Pain relief adequacy was converted into three numeric categories using SPSS syntax, with 1 representing No, 2 representing Uncertain, and 3 representing Yes. Among 118 patients, 85 (72.0%) reported adequate pain relief, 17 (14.4%) were uncertain, and 16 (13.6%) reported inadequate pain relief. The cross-tabulation confirmed that all patient responses were correctly assigned to their corresponding numeric codes. No missing values or coding mismatches were identified. The coded variable was prepared for further statistical analysis.

4.7 Coding Adverse Effects

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Did patient experienced any adverse effects * Postoperative Adverse Effects	118	100.0%	0	0.0%	118	100.0%

Did patient experienced any adverse effects * Postoperative Adverse Effects Crosstabulation

Count

		Postoperative Adverse Effects		
		No Adverse Effects	Adverse Effects Reported	Total
Did patient experienced any adverse effects	No	48	0	48
	Yes	0	70	70
Total		48	70	118

Figure 22: Numeric Coding and Cross-Validation of Postoperative Adverse-Effect Responses in IBM SPSS Statistics

We converted postoperative adverse-effect responses into numeric codes using SPSS syntax. Code 0 represented patients reporting no adverse effects, while code 1 represented patients reporting adverse effects. Among 118 patients, 70 (59.3%) reported one or more adverse effects, while 48 (40.7%) reported none. The cross-tabulation confirmed that all 48 negative responses were coded as 0 and all 70 positive responses as 1. No missing values or coding mismatches were identified. The coded variable was prepared for further statistical analysis.

4.8 Applying Compute Variable Formulas

IV Ketorolac Treatment Indicator

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	IV Paracetamol	59	50.0	50.0	50.0
	IV Ketorolac	59	50.0	50.0	100.0
	Total	118	100.0	100.0	

Severe or Worst Postoperative Pain

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Mild or Moderate Pain	73	61.9	61.9	61.9
	Severe or Worst Pain	45	38.1	38.1	100.0
	Total	118	100.0	100.0	

Positive Patient Satisfaction

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not Satisfied or Neutral	41	34.7	34.7	34.7
	Satisfied or Very Satisfied	77	65.3	65.3	100.0
	Total	118	100.0	100.0	

Figure 23: Frequency Distribution of Computed Treatment, Postoperative Pain Severity, and Patient Satisfaction Indicators in IBM SPSS Statistics

1. Treatment Group Indicator

Cleaned IV Treatment Group * IV Ketorolac Treatment Indicator Crosstabulation

Count

		IV Ketorolac Treatment Indicator		
		IV Paracetamol	IV Ketorolac	Total
Cleaned IV Treatment Group	IV Paracetamol	59	0	59
	IV Ketorolac	0	59	59
Total		59	59	118

2. High Pain Indicator

Postoperative VAS Pain Severity * Severe or Worst Postoperative Pain Crosstabulation

Count

		Severe or Worst Postoperative Pain		
		Mild or Moderate Pain	Severe or Worst Pain	Total
Postoperative VAS Pain Severity	Mild (1-3)	36	0	36
	Moderate (4-6)	37	0	37
	Severe (7-8)	0	41	41
	Worst Pain (9-10)	0	4	4
Total		73	45	118

3. Positive Satisfaction Indicator

Patient Satisfaction Level * Positive Patient Satisfaction Crosstabulation

Count

		Positive Patient Satisfaction		
		Not Satisfied or Neutral	Satisfied or Very Satisfied	Total
Patient Satisfaction Level	Very dissatisfied	6	0	6
	Dissatisfied	7	0	7
	Neutral	28	0	28
	Satisfied	0	37	37
	Very satisfied	0	40	40
Total		41	77	118

Figure 24: Cross-Validation of Computed Treatment, High Pain, and Positive Satisfaction Indicators Using IBM SPSS Statistics

Three computed variables were created in IBM SPSS Statistics to classify treatment groups, postoperative pain severity, and patient satisfaction. The treatment indicator identified 59 patients (50.0%) who received IV Paracetamol and 59 (50.0%) who received IV Ketorolac. The high pain indicator classified 73 patients (61.9%) as experiencing mild or moderate pain and 45 patients (38.1%) as experiencing severe or worst pain. The positive satisfaction indicator identified 77 patients (65.3%) who were satisfied or very satisfied, while 41 patients (34.7%) were classified as neutral or dissatisfied. We used crosstabulation to compare the computed variables with their original coded categories. All 118 patient records were correctly classified, with no missing values or coding mismatches. These calculations prepared the dataset for further statistical analysis.

4.9 Recode into Different Variables

Recoded Patient Satisfaction (3 Levels)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Dissatisfied	13	11.0	11.0	11.0
	Neutral	28	23.7	23.7	34.7
	Satisfied	77	65.3	65.3	100.0
	Total	118	100.0	100.0	

Recoded VAS Pain Severity (3 Levels)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Mild Pain	36	30.5	30.5	30.5
	Moderate Pain	37	31.4	31.4	61.9
	Severe or Worst Pain	45	38.1	38.1	100.0
	Total	118	100.0	100.0	

Patient Satisfaction Level * Recoded Patient Satisfaction (3 Levels) Crosstabulation

Count

		Recoded Patient Satisfaction (3 Levels)			
		Dissatisfied	Neutral	Satisfied	Total
Patient Satisfaction Level	Very dissatisfied	6	0	0	6
	Dissatisfied	7	0	0	7
	Neutral	0	28	0	28
	Satisfied	0	0	37	37
	Very satisfied	0	0	40	40
Total		13	28	77	118

Postoperative VAS Pain Severity * Recoded VAS Pain Severity (3 Levels) Crosstabulation

Count

		Recoded VAS Pain Severity (3 Levels)			
		Mild Pain	Moderate Pain	Severe or Worst Pain	Total
Postoperative VAS Pain Severity	Mild (1-3)	36	0	0	36
	Moderate (4-6)	0	37	0	37
	Severe (7-8)	0	0	41	41
	Worst Pain (9-10)	0	0	4	4
Total		36	37	45	118

Figure 25: Frequency Distribution and Cross-Validation of Recoded Patient Satisfaction and Postoperative VAS Pain Severity in IBM SPSS Statistics

We recoded patient satisfaction and postoperative VAS pain severity into three categories using IBM SPSS Statistics. Satisfaction results showed that 77 patients (65.3%) were satisfied, 28 (23.7%) were neutral, and 13 (11.0%) were dissatisfied. Pain severity results showed that 36 patients (30.5%) experienced mild pain, 37 (31.4%) experienced moderate pain, and 45 (38.1%) experienced severe or worst pain. Crosstabulation confirmed that all original satisfaction and pain categories were correctly assigned to their new groups. All 118 patient records were retained, with no missing values or recoding mismatches. The recoded variables were prepared for further statistical analysis.

4.10 Verification of Coded Data

Statistics

Final Verification of Coded Data

N	Valid	118
	Missing	0

Final Verification of Coded Data

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Coding Verified	118	100.0	100.0	100.0

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Cleaned IV Treatment Group * Final Verification of Coded Data	118	100.0%	0	0.0%	118	100.0%

Cleaned IV Treatment Group * Final Verification of Coded Data Crosstabulation

Count

		Final Verification of Coded Data	
		Coding Verified	Total
Cleaned IV Treatment Group	IV Paracetamol	59	59
	IV Ketorolac	59	59
Total		118	118

Figure 26: Final Verification of Numeric Codes, Computed Formulas, and Recoded Variables in IBM SPSS Statistics.

The coded dataset was checked in IBM SPSS Statistics to identify missing values, invalid numeric codes, and inconsistencies in computed and recoded variables. The verification process covered 14 variables, including demographic characteristics, clinical information, treatment groups, pain severity, patient satisfaction, pain relief adequacy, and adverse effects. Results confirmed that all 118 pa-

tient records (100.0%) passed the specified coding checks, with no missing values or detected coding mismatches. Treatment group verification identified 59 patients receiving IV Paracetamol and 59 receiving IV Ketorolac, with all records successfully verified. These findings confirmed that the coded variables were consistent with the applied formulas and suitable for further statistical analysis. Clinical eligibility and the accuracy of original patient records require separate verification.

5. Descriptive Statistical Analysis

5.1 Frequencies and Percentages

Patient Age Group

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
18–25 years	49	41.5	41.5	41.5
26–40 years	30	25.4	25.4	66.9
41–60 years	30	25.4	25.4	92.4
Above 60 years	9	7.6	7.6	100.0
Total	**118**	**100.0**	**100.0**	–

Patient Gender

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
Female	50	42.4	42.4	42.4
Male	68	57.6	57.6	100.0
Total	**118**	**100.0**	**100.0**	–

Patient BMI Category

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
Less than 18.5	9	7.6	7.6	7.6
18.5 – 24.9	57	48.3	48.3	55.9
25.0 – 29.9	34	28.8	28.8	84.7
30.0 – 39.9	13	11.0	11.0	95.8
Above 40	5	4.2	4.2	100.0
Total	**118**	**100.0**	**100.0**	–

Patient ASA Physical Status

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
ASA 1	74	62.7	62.7	62.7
ASA 2	27	22.9	22.9	85.6
ASA 3	15	12.7	12.7	98.3
ASA 4	2	1.7	1.7	100.0
Total	**118**	**100.0**	**100.0**	–

Cleaned IV Treatment Group

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
IV Paracetamol	59	50.0	50.0	50.0
IV Ketorolac	59	50.0	50.0	100.0
Total	**118**	**100.0**	**100.0**	–

Postoperative VAS Pain Severity

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
Mild (1–3)	36	30.5	30.5	30.5
Moderate (4–6)	37	31.4	31.4	61.9
Severe (7–8)	41	34.7	34.7	96.6
Worst Pain (9–10)	4	3.4	3.4	100.0
Total	**118**	**100.0**	**100.0**	–

Patient Satisfaction Level

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
Very dissatisfied	6	5.1	5.1	5.1
Dissatisfied	7	5.9	5.9	11.0
Neutral	28	23.7	23.7	34.7
Satisfied	37	31.4	31.4	66.1
Very satisfied	40	33.9	33.9	100.0
Total	**118**	**100.0**	**100.0**	–

Adequacy of Postoperative Pain Relief

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
No	16	13.6	13.6	13.6
Uncertain	17	14.4	14.4	28.0
Yes	85	72.0	72.0	100.0
Total	**118**	**100.0**	**100.0**	–

Postoperative Adverse Effects

Category	Frequency (n)	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
No Adverse Effects	48	40.7	40.7	40.7
Adverse Effects Reported	70	59.3	59.3	100.0
Total	**118**	**100.0**	**100.0**	–

Figure 27: Frequency and Percentage Distribution of Patient Demographics, Clinical Characteristics, Treatment Groups, and Postoperative Outcomes in IBM SPSS Statistics.

We performed frequency and percentage analyses on 118 patient records using IBM SPSS Statistics. The largest age group was 18–25 years (41.5%), and 57.6% of patients were male. Most patients had a BMI of 18.5–24.9 (48.3%) and ASA I status (62.7%). Both treatment groups contained 59 patients (50.0% each). Severe postoperative pain was reported by 34.7% of patients, while 65.3% were satisfied or very satisfied with treatment. Adequate pain relief was reported by 72.0%, and 59.3% experienced adverse effects. All nine variables contained 118 valid responses.

5.2 Age and Gender Distribution

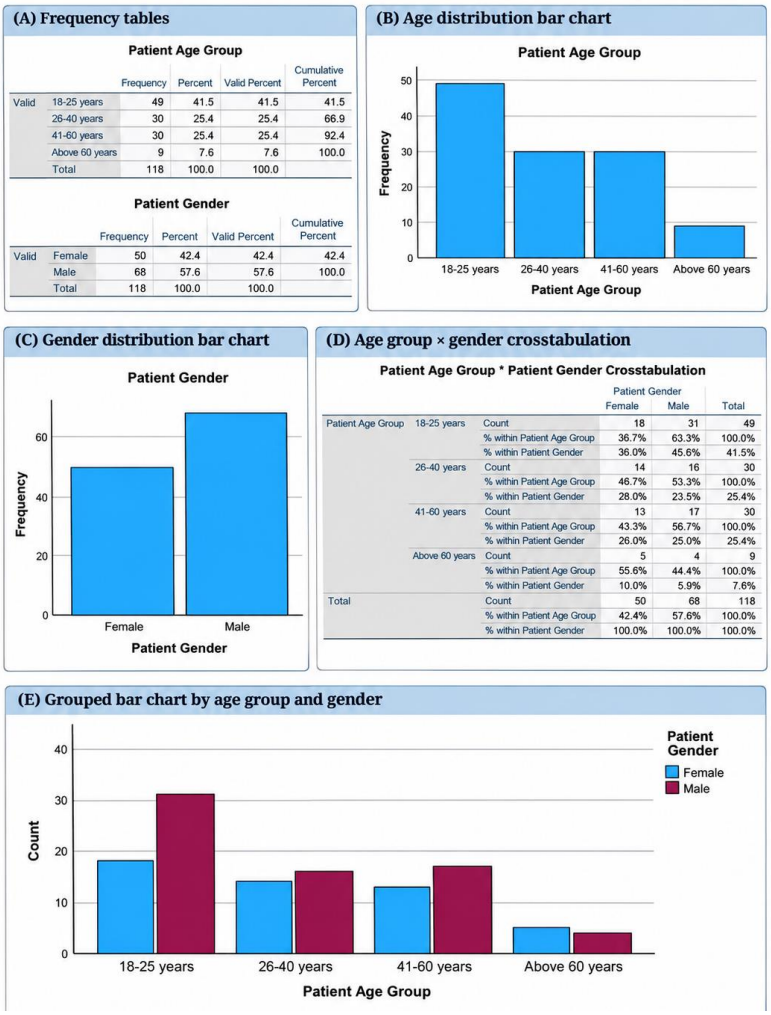


Figure 28: Frequency Distribution, Bar Charts, and Crosstabulation of Patient Age Groups and Gender in IBM SPSS Statistics (N = 118).

We analysed age and gender distribution using IBM SPSS Statistics for 118 patients. The largest age group was 18–25 years, with 49 patients (41.5%). The 26–40 and 41–60 age groups each included 30 patients (25.4%), while 9 patients (7.6%) were above 60 years. Gender distribution showed 68 males (57.6%) and 50 females (42.4%).

Crosstabulation showed that the 18–25 age group included 31 males and 18 females. The 26–40 age group included 16 males and 14 females, while the 41–60 age group included 17 males and 13 females. Among patients above 60 years, 5 were female, and 4 were male. The grouped bar chart confirmed that males outnumbered females in the first three age groups. All 118 patient records were included in the analysis.

5.3 BMI and ASA Status



Figure 29: Frequency Distribution, Bar Charts, and Crosstabulation of Patient BMI Categories and ASA Physical Status in IBM SPSS Statistics (N = 118).

We analysed BMI and ASA physical status for 118 patients. The most common BMI category was 18.5–24.9 (57 patients, 48.3%), followed by 25.0–29.9 (34 patients, 28.8%). ASA I was the most frequent classification, representing 74 patients (62.7%), followed by ASA II with 27 patients (22.9%).

Crosstabulation showed that 45 patients in the 18.5–24.9 BMI category had ASA I status. In the 25.0–29.9 BMI category, 17 patients had ASA I and 11 had ASA II status. The grouped bar chart confirmed that ASA I was the most common status in the lower BMI categories. All 118 records were included in the analysis.

5.4 Distribution of Treatment Groups

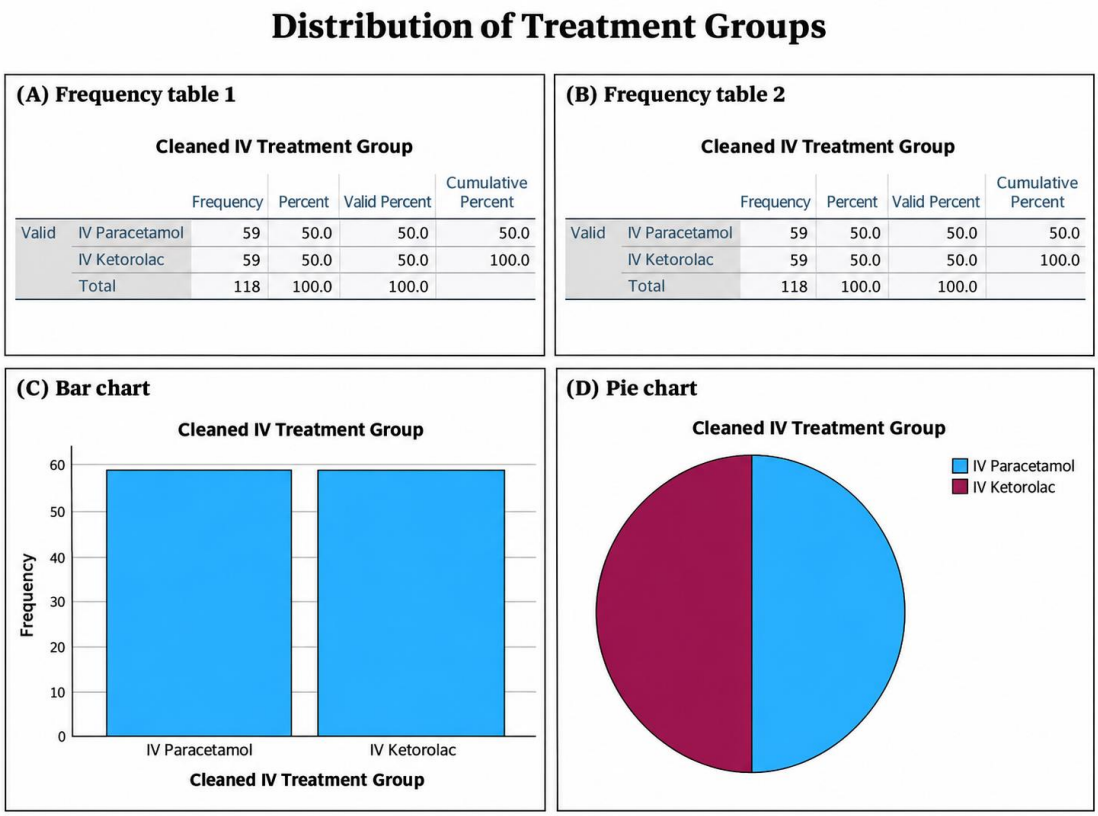


Figure 30: Frequency Distribution, Bar Chart, and Pie Chart of IV Paracetamol and IV Ketorolac Treatment Groups in IBM SPSS Statistics (N = 118).

We analysed the distribution of treatment groups using IBM SPSS Statistics. Of the 118 patients, 59 (50.0%) received IV Paracetamol, and the remaining 59 (50.0%) received IV Ketorolac. Both frequency tables confirmed an equal distribution of patients between the two treatment groups. The bar chart showed identical patient counts, and the pie chart displayed an equal 50.0% proportion for each group. All 118 patient records were included, with no missing treatment group data. This equal group distribution provides a balanced sample size for comparing postoperative pain outcomes between the two medications.

5.5 VAS Pain Severity



Figure 31: Frequency Distributions, Bar Chart, Pie Chart, Crosstabulation, and Treatment Group Comparison of Postoperative VAS Pain Severity in IBM SPSS Statistics (N = 118).

We analysed postoperative pain severity using the Visual Analogue Scale (VAS) among 118 patients. Forty-one patients (34.7%) reported severe pain (VAS 7–8), followed by 37 patients (31.4%) with moderate pain (VAS 4–6) and 36 patients (30.5%) with mild pain (VAS 1–3). Only 4 patients (3.4%) reported worst pain (VAS 9–10).

Crosstabulation showed similar pain severity distributions between the two treatment groups. In the IV Paracetamol group, 18 patients reported mild pain, 19 moderate pain, 20 severe pain, and 2 worst pain. The IV Ketorolac group included 18 patients with mild pain, 18 with moderate pain, 21 with severe pain, and 2 with the worst pain. The grouped bar chart confirmed that the observed differences were small. Both groups contained 59 patients, with no missing VAS data. Statistical testing is required to determine whether the differences are significant.

5.6 Patient Satisfaction



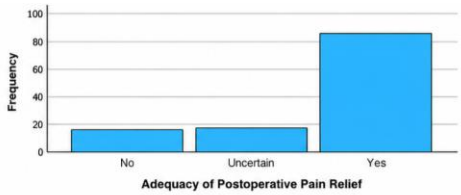
Figure 32: Grouped Bar Chart Comparing Patient Satisfaction Levels Between IV Paracetamol and IV Ketorolac Groups (N = 118).

Patient satisfaction differed between the two treatment groups. In the IV Paracetamol group, 31 patients (52.5%) were very satisfied, compared with 9 patients (15.3%) in the IV Ketorolac group. A further 16 patients (27.1%) in the Paracetamol group and 21 patients (35.6%) in the Ketorolac group reported being satisfied. Neutral responses were more common in the Ketorolac group (33.9%) than in the Paracetamol group (13.6%). Dissatisfaction was also higher in the Ketorolac group (10.2%) compared with the Paracetamol group (1.7%). Three patients (5.1%) in each group were very dissatisfied. The results show higher reported satisfaction in the IV Paracetamol group, although inferential testing is needed to confirm statistical significance.

5.7 Adequacy of Pain Relief

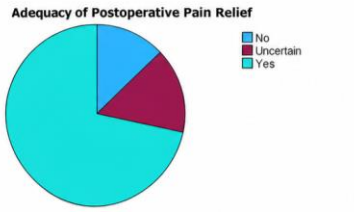
Adequacy of Postoperative Pain Relief

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	16	13.6	13.6	13.6
	Uncertain	17	14.4	14.4	28.0
	Yes	85	72.0	72.0	100.0
	Total	118	100.0	100.0	



Adequacy of Postoperative Pain Relief

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	16	13.6	13.6	13.6
	Uncertain	17	14.4	14.4	28.0
	Yes	85	72.0	72.0	100.0
	Total	118	100.0	100.0	



Cleaned IV Treatment Group * Adequacy of Postoperative Pain Relief Crosstabulation

			Adequacy of Postoperative Pain Relief			
			No	Uncertain	Yes	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	5	5	49	59
		% within Cleaned IV Treatment Group	8.5%	8.5%	83.1%	100.0%
		% within Adequacy of Postoperative Pain Relief	31.3%	29.4%	57.6%	50.0%
	IV Ketorolac	Count	11	12	36	59
		% within Cleaned IV Treatment Group	18.6%	20.3%	61.0%	100.0%
		% within Adequacy of Postoperative Pain Relief	68.8%	70.6%	42.4%	50.0%
Total		Count	16	17	85	118
		% within Cleaned IV Treatment Group	13.6%	14.4%	72.0%	100.0%
		% within Adequacy of Postoperative Pain Relief	100.0%	100.0%	100.0%	100.0%

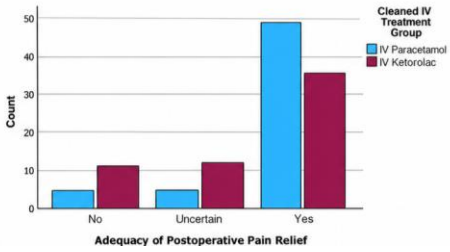


Figure 33: Frequency Distribution, Bar Chart, Pie Chart, Crosstabulation, and Treatment Group Comparison of Adequacy of Postoperative Pain Relief (N = 118).

Of 118 patients, 85 (72.0%) reported adequate postoperative pain relief, 16 (13.6%) reported inadequate relief, and 17 (14.4%) were uncertain. In the IV Paracetamol group, 49 of 59 patients (83.1%) reported adequate pain relief, compared with 36 of 59 patients (61.0%) in the IV Ketorolac group. Inadequate pain relief was reported by 5 patients (8.5%) in the Paracetamol group and 11 patients (18.6%) in the Ketorolac group. Similarly, uncertainty was higher in the Ketorolac group (20.3%) than in the Paracetamol group (8.5%). These descriptive findings indicate that patients receiving IV Paracetamol reported adequate pain relief more frequently than those receiving IV Ketorolac. Inferential analysis will assess statistical significance.

5.8 Adverse Effects and Their Types

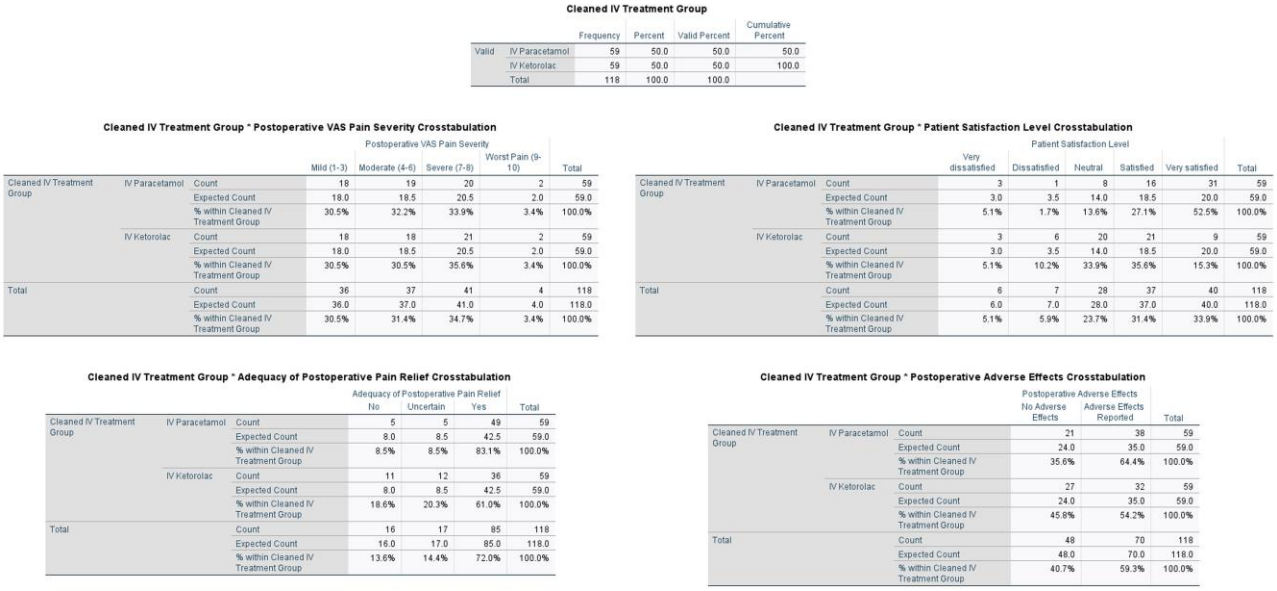


Figure 34: Frequency Tables, Bar Chart, Pie Chart, Treatment Group Crosstabulation, Grouped Bar Chart, and Distribution of Recorded Postoperative Adverse Effect Types Among Patients Receiving IV Paracetamol and IV Ketorolac (N = 118).

Among 118 patients, 70 (59.3%) reported postoperative adverse effects, while 48 (40.7%) reported none. In the IV Paracetamol group, 38 of 59 patients (64.4%) experienced adverse effects, compared with 32 of 59 patients (54.2%) in the IV Ketorolac group. The most frequently reported adverse effects were nausea with vomiting (10 patients, 14.3% of those reporting adverse effects), GI discomfort (9 patients, 12.9%), dizziness (6 patients, 8.6%), and dizziness with GI discomfort (6 patients, 8.6%). Other recorded effects included allergic reactions, breathing difficulties, bleeding complications, and restlessness, sometimes occurring together. These descriptive findings indicate a higher reported frequency of adverse effects in the IV Paracetamol group. However, the difference between treatment groups requires statistical testing to determine significance.

6. Inferential Statistical Analysis

6.1 Selection and Assumptions of Statistical Tests



Cleaned IV Treatment Group

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	IV Paracetamol	59	50.0	50.0	50.0
	IV Ketorolac	59	50.0	50.0	100.0
	Total	118	100.0	100.0	

Cleaned IV Treatment Group * Postoperative VAS Pain Severity Crosstabulation

			Postoperative VAS Pain Severity				
			Mild (1-3)	Moderate (4-6)	Severe (7-8)	Worst Pain (9-10)	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	18	19	20	2	59
		Expected Count	18.0	18.5	20.5	2.0	59.0
		% within Cleaned IV Treatment Group	30.5%	32.2%	33.9%	3.4%	100.0%
	IV Ketorolac	Count	18	18	21	2	59
		Expected Count	18.0	18.5	20.5	2.0	59.0
		% within Cleaned IV Treatment Group	30.5%	30.5%	35.6%	3.4%	100.0%
Total		Count	36	37	41	4	118
		Expected Count	36.0	37.0	41.0	4.0	118.0
		% within Cleaned IV Treatment Group	30.5%	31.4%	34.7%	3.4%	100.0%

Cleaned IV Treatment Group * Patient Satisfaction Level Crosstabulation

			Patient Satisfaction Level					
			Very dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	3	1	8	16	31	59
		Expected Count	3.0	3.5	14.0	18.5	20.0	59.0
		% within Cleaned IV Treatment Group	5.1%	1.7%	13.6%	27.1%	52.5%	100.0%
	IV Ketorolac	Count	3	6	20	21	9	59
		Expected Count	3.0	3.5	14.0	18.5	20.0	59.0
		% within Cleaned IV Treatment Group	5.1%	10.2%	33.9%	35.6%	15.3%	100.0%
Total		Count	6	7	28	37	40	118
		Expected Count	6.0	7.0	28.0	37.0	40.0	118.0
		% within Cleaned IV Treatment Group	5.1%	5.9%	23.7%	31.4%	33.9%	100.0%

Cleaned IV Treatment Group * Adequacy of Postoperative Pain Relief Crosstabulation

			Adequacy of Postoperative Pain Relief			
			No	Uncertain	Yes	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	5	5	49	59
		Expected Count	8.0	8.5	42.5	59.0
		% within Cleaned IV Treatment Group	8.5%	8.5%	83.1%	100.0%
	IV Ketorolac	Count	11	12	36	59
		Expected Count	8.0	8.5	42.5	59.0
		% within Cleaned IV Treatment Group	18.6%	20.3%	61.0%	100.0%
Total		Count	16	17	85	118
		Expected Count	16.0	17.0	85.0	118.0
		% within Cleaned IV Treatment Group	13.6%	14.4%	72.0%	100.0%

Cleaned IV Treatment Group * Postoperative Adverse Effects Crosstabulation

			Postoperative Adverse Effects		
			No Adverse Effects	Adverse Effects Reported	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	21	38	59
		Expected Count	24.0	35.0	59.0
		% within Cleaned IV Treatment Group	35.6%	64.4%	100.0%
	IV Ketorolac	Count	27	32	59
		Expected Count	24.0	35.0	59.0
		% within Cleaned IV Treatment Group	45.8%	54.2%	100.0%
Total		Count	48	70	118
		Expected Count	48.0	70.0	118.0
		% within Cleaned IV Treatment Group	40.7%	59.3%	100.0%

Figure 35: Verification of Treatment Group Distribution and Expected Cell Frequencies for Postoperative Pain Severity, Patient Satisfaction, Adequacy of Pain Relief, and Adverse Effects in IBM SPSS Statistics (N = 118).

The analysis included 118 patients, equally distributed between IV Paracetamol (n = 59) and IV Ketorolac (n = 59). We examined expected cell counts to assess the assumptions of Pearson's Chi-Square Test. The VAS pain severity table contained two expected counts below five, representing 25% of its cells. The patient satisfaction table contained four expected counts below five, representing 40% of its cells. These results indicate that the expected-frequency assumption was not fully satisfied for these comparisons. All expected counts were at least five for adequacy of pain relief and adverse effects. We therefore used exact tests for tables with low expected counts.

6.2 Chi-Square Test for VAS Pain Severity

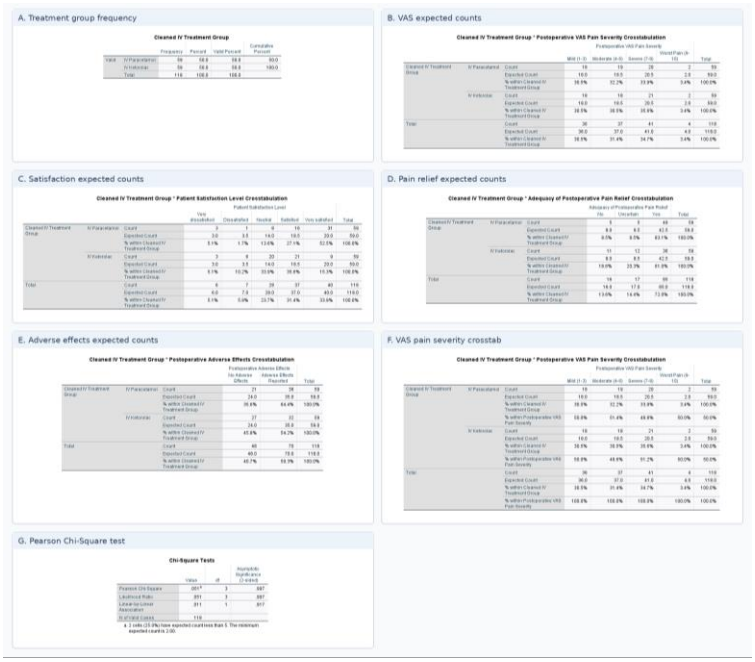


Figure 36: Pearson Chi-Square Test Results for Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Pearson Chi-Square Test showed no statistically significant association between treatment group and postoperative VAS pain severity, $\chi^2(3, N = 118) = 0.051$, $p = 0.997$. Pain severity distributions were similar in both groups. However, two cells (25%) had expected counts below five, so the result requires confirmation through an exact test.

6.3 Fisher's Exact Test

A. VAS pain severity crosstabulation

Cleaned IV Treatment Group * Postoperative VAS Pain Severity Crosstabulation

			Postoperative VAS Pain Severity				
			Mild (1-3)	Moderate (4-6)	Severe (7-8)	Worst Pain (9-10)	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	18	19	20	2	59
		Expected Count	18.0	18.5	20.5	2.0	59.0
		% within Cleaned IV Treatment Group	30.5%	32.2%	33.9%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	50.0%	51.4%	48.8%	50.0%	50.0%
	IV Ketorolac	Count	18	18	21	2	59
		Expected Count	18.0	18.5	20.5	2.0	59.0
		% within Cleaned IV Treatment Group	30.5%	30.5%	35.6%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	50.0%	48.6%	51.2%	50.0%	50.0%
Total		Count	36	37	41	4	118
		Expected Count	36.0	37.0	41.0	4.0	118.0
		% within Cleaned IV Treatment Group	30.5%	31.4%	34.7%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	100.0%	100.0%	100.0%	100.0%	100.0%

B. Chi-Square and Fisher-Freeman-Halton Exact Tests

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)	Point Probability
Pearson Chi-Square	.051[a]	3	.997	1.000		
Likelihood Ratio	.051	3	.997	1.000		
Fisher-Freeman-Halton Exact Test	.209			1.000		
Linear-by-Linear Association	.011[b]	1	.917	1.000	.500	.082
N of Valid Cases	118					

a. 2 cells (25.0%) have expected count less than 5. The minimum expected count is 2.00.
b. The standardized statistic is .104.

Figure 37: Fisher-Freeman-Halton Exact Test for Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Fisher-Freeman-Halton Exact Test showed no statistically significant association between treatment group and postoperative VAS pain severity (exact $p = 1.000$). The findings support the earlier Pearson Chi-Square result ($p = 0.997$). Therefore, we found no significant difference in the distribution of VAS pain severity between the two treatment groups.

6.4 Mann-Whitney U Test

A. Ranks

Ranks

	Cleaned IV Treatment Group	N	Mean Rank	Sum of Ranks
Postoperative VAS Pain Severity	IV Paracetamol	59	59.17	3491.00
	IV Ketorolac	59	59.83	3530.00
	Total	118		

B. Test Statistics

Test Statistics[a]

	Postoperative VAS Pain Severity
Mann-Whitney U	1721.000
Wilcoxon W	3491.000
Z	-.111
Asymp. Sig. (2-tailed)	.912

a. Grouping Variable: Cleaned IV Treatment Group

Figure 38: Mann-Whitney U Test Comparing Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Mann-Whitney U Test showed no statistically significant difference in postoperative VAS pain severity between the IV Paracetamol and IV Ketorolac groups (U = 1721.000, Z = -0.111, p = 0.912). The mean ranks were 59.17 for IV Paracetamol and 59.83 for IV Ketorolac, indicating very similar pain severity distributions. Because the p-value was greater than 0.05, we did not reject the null hypothesis.

6.5 Chi-Square Test for Patient Satisfaction

Cleaned IV Treatment Group * Patient Satisfaction Level Crosstabulation

			Patient Satisfaction Level					
			Very dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	3	1	8	16	31	59
		Expected Count	3.0	3.5	14.0	18.5	20.0	59.0
		% within Cleaned IV Treatment Group	5.1%	1.7%	13.6%	27.1%	52.5%	100.0%
		% within Patient Satisfaction Level	50.0%	14.3%	28.6%	43.2%	77.5%	50.0%
	IV Ketorolac	Count	3	6	20	21	9	59
		Expected Count	3.0	3.5	14.0	18.5	20.0	59.0
		% within Cleaned IV Treatment Group	5.1%	10.2%	33.9%	35.6%	15.3%	100.0%
		% within Patient Satisfaction Level	50.0%	85.7%	71.4%	56.8%	22.5%	50.0%
Total		Count	6	7	28	37	40	118
		Expected Count	6.0	7.0	28.0	37.0	40.0	118.0
		% within Cleaned IV Treatment Group	5.1%	5.9%	23.7%	31.4%	33.9%	100.0%
		% within Patient Satisfaction Level	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)	Point Probability
Pearson Chi-Square	21.490[a]	4	<.001	<.001		
Likelihood Ratio	22.752	4	<.001	<.001		
Fisher-Freeman-Halton Exact Test	21.758			<.001		
Linear-by-Linear Association	13.093[b]	1	<.001	<.001	<.001	.000
N of Valid Cases	118					

a. 4 cells (40.0%) have expected count less than 5. The minimum expected count is 3.00.

b. The standardized statistic is -3.618.

Figure 39: Pearson Chi-Square and Fisher-Freeman-Halton Exact Test Results for Patient Satisfaction Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Pearson Chi-Square Test showed a statistically significant association between treatment group and patient satisfaction, $\chi^2(4, N = 118) = 21.490$, $p < 0.001$. The Fisher-Freeman-Halton Exact Test confirmed this finding ($p < 0.001$). Very high satisfaction was reported by 52.5% of patients receiving IV Paracetamol compared with 15.3% receiving IV Ketorolac. These findings indicate higher patient satisfaction in the IV Paracetamol group.

6.6 Chi-Square Test for Adequate Pain Relief

Cleaned IV Treatment Group * Adequacy of Postoperative Pain Relief Crosstabulation

			Adequacy of Postoperative Pain Relief			
			No	Uncertain	Yes	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	5	5	49	59
		Expected Count	8.0	8.5	42.5	59.0
		% within Cleaned IV Treatment Group	8.5%	8.5%	83.1%	100.0%
		% within Adequacy of Postoperative Pain Relief	31.3%	29.4%	57.6%	50.0%
	IV Ketorolac	Count	11	12	36	59
		Expected Count	8.0	8.5	42.5	59.0
		% within Cleaned IV Treatment Group	18.6%	20.3%	61.0%	100.0%
		% within Adequacy of Postoperative Pain Relief	68.8%	70.6%	42.4%	50.0%
Total		Count	16	17	85	118
		Expected Count	16.0	17.0	85.0	118.0
		% within Cleaned IV Treatment Group	13.6%	14.4%	72.0%	100.0%
		% within Adequacy of Postoperative Pain Relief	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	7.121[a]	2	.028
Likelihood Ratio	7.272	2	.026
Linear-by-Linear Association	5.901	1	.015
N of Valid Cases	118		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 8.00.

Figure 40: Pearson Chi-Square Test Comparing Adequacy of Postoperative Pain Relief Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Pearson Chi-Square Test showed a statistically significant association between treatment group and adequacy of postoperative pain relief, $\chi^2(2, N = 118) = 7.121$, $p = 0.028$. Adequate pain relief was reported by 83.1% of patients receiving IV Paracetamol, compared with 61.0% receiving IV Ketorolac. All expected cell counts were above five, satisfying the Chi-Square expected-frequency assumption. The results indicate that the IV Paracetamol group reported adequate pain relief more frequently.

6.7 Chi-Square Test for Adverse Effects

Cleaned IV Treatment Group * Postoperative Adverse Effects Crosstabulation

			Postoperative Adverse Effects		
			No Adverse Effects	Adverse Effects Reported	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	21	38	59
		Expected Count	24.0	35.0	59.0
		% within Cleaned IV Treatment Group	35.6%	64.4%	100.0%
		% within Postoperative Adverse Effects	43.8%	54.3%	50.0%
	IV Ketorolac	Count	27	32	59
		Expected Count	24.0	35.0	59.0
		% within Cleaned IV Treatment Group	45.8%	54.2%	100.0%
		% within Postoperative Adverse Effects	56.3%	45.7%	50.0%
Total		Count	48	70	118
		Expected Count	48.0	70.0	118.0
		% within Cleaned IV Treatment Group	40.7%	59.3%	100.0%
		% within Postoperative Adverse Effects	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	1.264[a]	1	.261		
Continuity Correction[b]	.878	1	.349		
Likelihood Ratio	1.267	1	.260		
Fisher's Exact Test				.349	.174
Linear-by-Linear Association	1.254	1	.263		
N of Valid Cases	118				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 24.00.

b. Computed only for a 2x2 table

Figure 41: Pearson Chi-Square Test Results for Postoperative Adverse Effects Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The Pearson Chi-Square Test showed no statistically significant association between treatment group and postoperative adverse effects, $\chi^2(1, N = 118) = 1.264$, $p = 0.261$. Adverse effects were reported

by 64.4% of patients receiving IV Paracetamol and 54.2% receiving IV Ketorolac. All expected cell counts were above five. Although adverse effects were reported more frequently in the IV Paracetamol group, the difference was not statistically significant.

6.8 P-Values and Statistical Significance

Table 1: Statistical Significance of Postoperative Outcomes Between IV Paracetamol and IV Ketorolac Groups (N = 118).

Statistical test	p-value	Result
Pearson Chi-Square, VAS Pain	0.997	Not significant
Fisher-Freeman-Halton, VAS Pain	1.000	Not significant
Mann-Whitney U, VAS Pain	0.912	Not significant
Chi-Square, Patient Satisfaction	< 0.001	Significant
Chi-Square, Adequate Pain Relief	0.028	Significant
Chi-Square, Adverse Effects	0.261	Not significant

We assessed statistical significance at $p < 0.05$. The results showed significant associations between treatment group and patient satisfaction ($p < 0.001$) and adequacy of postoperative pain relief ($p = 0.028$). Patients receiving IV Paracetamol reported higher satisfaction and more adequate pain relief than those receiving IV Ketorolac. However, VAS pain severity did not differ significantly between treatment groups by Pearson's Chi-Square Test ($p = 0.997$), Fisher-Freeman-Halton Exact Test ($p = 1.000$), or Mann-Whitney U Test ($p = 0.912$). Postoperative adverse effects also showed no significant association with treatment group ($p = 0.261$). The reported significance levels were not adjusted for multiple comparisons.

6.9 Summary of Inferential Results

Inferential statistical analysis was performed on 118 patient records: 59 in the IV Paracetamol group and 59 in the IV Ketorolac group. The analysis examined differences in postoperative pain severity, patient satisfaction, adequacy of pain relief, and adverse effects using Pearson's Chi-Square Test, Fisher's Exact Test, and the Mann-Whitney U Test.

Postoperative VAS pain severity showed no statistically significant association with treatment group, $\chi^2(3) = 0.051$, $p = 0.997$. The Fisher-Freeman-Halton Exact Test supported this result ($p = 1.000$). Similarly, the Mann-Whitney U Test found no significant difference in ordinal pain severity between the groups ($U = 1721.000$, $p = 0.912$).

Patient satisfaction showed a statistically significant association with treatment group, $\chi^2(4) = 21.490$, $p < 0.001$, which was confirmed by exact testing. Very high satisfaction was reported by 52.5% of patients receiving IV Paracetamol compared with 15.3% receiving IV Ketorolac. Adequacy of pain relief also showed a significant association, $\chi^2(2) = 7.121$, $p = 0.028$. Adequate relief was reported by 83.1% of the Paracetamol group and 61.0% of the Ketorolac group.

Postoperative adverse effects showed no statistically significant association with treatment group, $\chi^2(1) = 1.264$, $p = 0.261$, despite higher reported frequencies in the IV Paracetamol group (64.4%) than in the IV Ketorolac group (54.2%).

The findings indicate higher reported patient satisfaction and adequacy of pain relief in the IV Paracetamol group. However, we did not detect statistically significant differences in VAS pain severity

or adverse effects. These results require cautious interpretation because multiple comparisons were performed without adjustment, and previously identified patient eligibility concerns remain subject to clinical verification.

7. Graphical Presentation in SPSS

7.1 Pie Chart of Patient Satisfaction

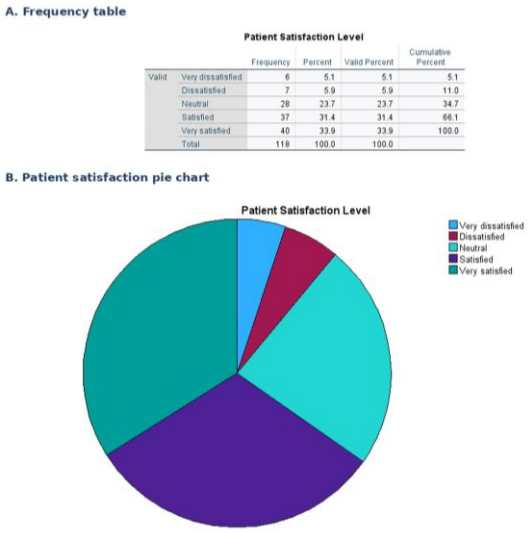


Figure 42: Frequency Distribution and Pie Chart of Patient Satisfaction Levels Following Postoperative Pain Management (N = 118).

The pie chart shows that 40 patients (33.9%) were very satisfied and 37 (31.4%) were satisfied with postoperative pain management. Another 28 patients (23.7%) reported neutral satisfaction, while 7 (5.9%) were dissatisfied and 6 (5.1%) were very dissatisfied. A total of 77 patients (65.3%) reported positive satisfaction, indicating that most patients were satisfied with their postoperative pain management.

7.2 Bar Chart of VAS Pain Severity

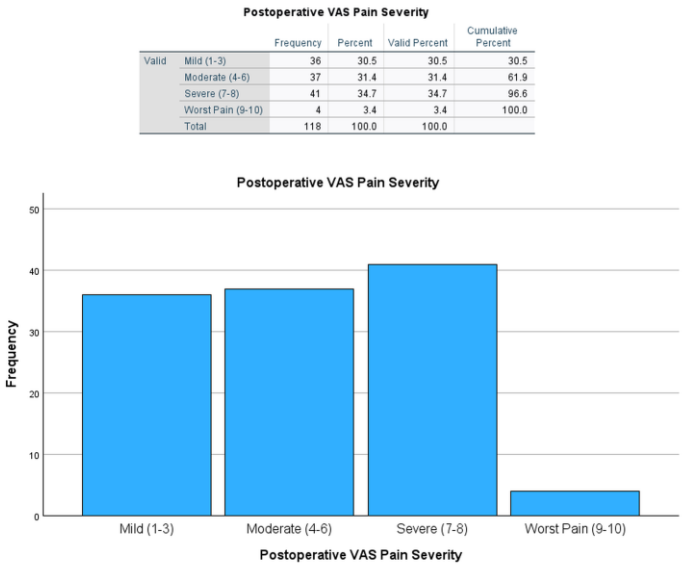


Figure 43: Frequency Distribution and Bar Chart of Postoperative VAS Pain Severity Among 118 Patients.

Among 118 patients, 41 (34.7%) experienced severe postoperative pain, followed by 37 (31.4%) with moderate pain and 36 (30.5%) with mild pain. Only 4 patients (3.4%) reported worst pain. Se-

vere pain was the most frequently reported category. A total of 45 patients (38.1%) experienced severe or worst pain.

7.3 Bar Chart of Age Distribution

A. Patient age group frequency table

Patient Age Group

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-25 years	49	41.5	41.5	41.5
	26-40 years	30	25.4	25.4	66.9
	41-60 years	30	25.4	25.4	92.4
	Above 60 years	9	7.6	7.6	100.0
	Total	118	100.0	100.0	

B. Patient age distribution bar chart

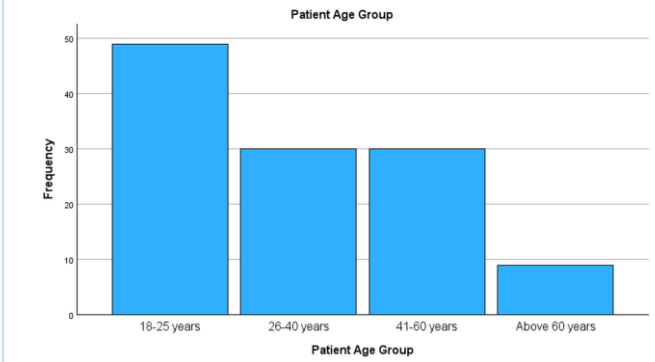


Figure 44: Frequency Distribution and Bar Chart of Patient Age Groups Among 118 Patients.

Among 118 patients, the largest age group was 18–25 years, with 49 patients (41.5%). The 26–40 and 41–60 age groups each included 30 patients (25.4%). Only 9 patients (7.6%) were above 60 years. The findings show that younger patients represented the largest proportion of the study records.

7.4 Bar Chart of BMI Categories

A. Patient BMI Category Frequency Table

Patient BMI Category

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Less than 18.5	9	7.6	7.6	7.6
	18.5 - 24.9	57	48.3	48.3	55.9
	25.0 - 29.9	34	28.8	28.8	84.7
	30.0 - 39.9	13	11.0	11.0	95.8
	Above 40	5	4.2	4.2	100.0
	Total	118	100.0	100.0	

B. Patient BMI Category Bar Chart

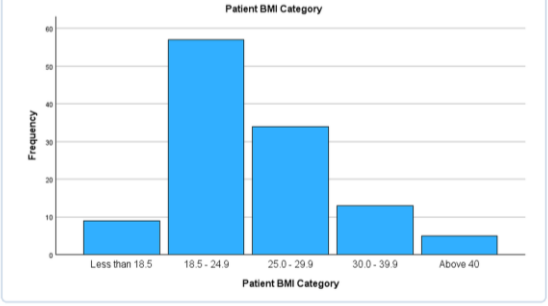


Figure 45: Frequency Distribution and Bar Chart of Patient BMI Categories Among 118 Patients.

Among 118 patients, the largest BMI category was 18.5–24.9, with 57 patients (48.3%), followed by 25.0–29.9 with 34 patients (28.8%). Another 13 patients (11.0%) had a BMI of 30.0–39.9, while 9 (7.6%) had a BMI below 18.5. Only 5 patients (4.2%) had a BMI above 40. The findings show that nearly half of patients fell into the 18.5–24.9 BMI category.

7.5 Bar Chart of Adequate Pain Relief

7.5 Bar Chart of Adequate Pain Relief

A. Frequency table

Adequacy of Postoperative Pain Relief

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	16	13.6	13.6	13.6
	Uncertain	17	14.4	14.4	28.0
	Yes	85	72.0	72.0	100.0
	Total	118	100.0	100.0	

B. Bar chart

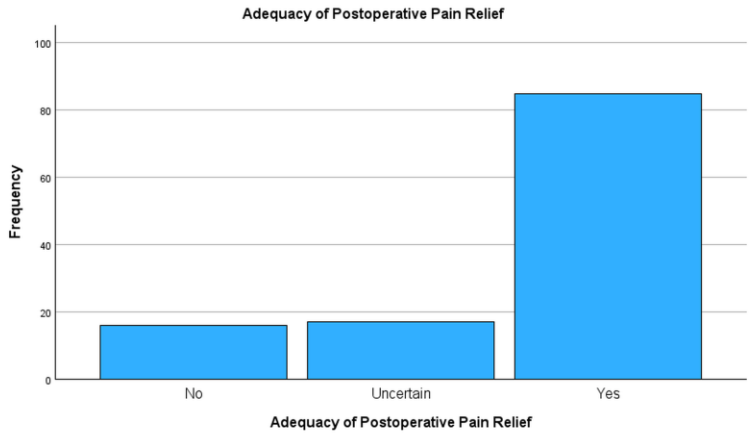


Figure 46: Frequency Distribution and Bar Chart of Adequacy of Postoperative Pain Relief Among 118 Patients.

Of the 118 patients, 85 (72.0%) reported adequate postoperative pain relief, 17 (14.4%) were uncertain, and 16 (13.6%) reported inadequate relief. The bar chart shows that adequate pain relief was the most frequently reported response, indicating that most patients considered their postoperative pain management adequate.

7.6 Grouped Bar Chart Comparing Paracetamol and Ketorolac

A. Treatment group × VAS pain severity crosstabulation

Cleaned IV Treatment Group * Postoperative VAS Pain Severity Crosstabulation

			Postoperative VAS Pain Severity				
			Mild (1-3)	Moderate (4-6)	Severe (7-8)	Worst Pain (9-10)	Total
Cleaned IV Treatment Group	IV Paracetamol	Count	18	19	20	2	59
		% within Cleaned IV Treatment Group	30.5%	32.2%	33.9%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	50.0%	51.4%	48.8%	50.0%	50.0%
	IV Ketorolac	Count	18	18	21	2	59
		% within Cleaned IV Treatment Group	30.5%	30.5%	35.6%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	50.0%	48.6%	51.2%	50.0%	50.0%
Total		Count	36	37	41	4	118
		% within Cleaned IV Treatment Group	30.5%	31.4%	34.7%	3.4%	100.0%
		% within Postoperative VAS Pain Severity	100.0%	100.0%	100.0%	100.0%	100.0%

B. Grouped bar chart comparing IV Paracetamol and IV Ketorolac

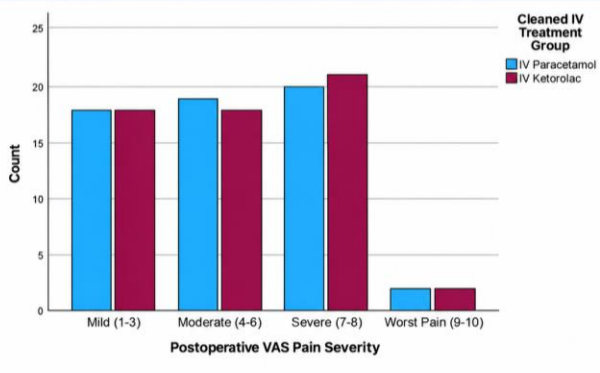


Figure 47: Grouped Bar Chart and Crosstabulation Comparing Postoperative VAS Pain Severity Between IV Paracetamol and IV Ketorolac Groups (N = 118).

The grouped bar chart shows similar postoperative pain severity distributions in both treatment groups. In each group, 18 patients (30.5%) reported mild pain. Moderate pain occurred in 19 patients (32.2%) receiving IV Paracetamol and 18 (30.5%) receiving IV Ketorolac. Severe pain was reported by 20 (33.9%) and 21 (35.6%) patients, respectively. Two patients (3.4%) in each group reported the

worst pain. The earlier statistical analysis confirmed no significant difference in VAS pain severity between the treatment groups (exact p = 1.000).

7.7 Bar Chart of Adverse Effects

A. Postoperative adverse effects frequency table

Postoperative Adverse Effects

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No Adverse Effects	48	40.7	40.7	40.7
	Adverse Effects Reported	70	59.3	59.3	100.0
	Total	118	100.0	100.0	

B. Postoperative adverse effects bar chart

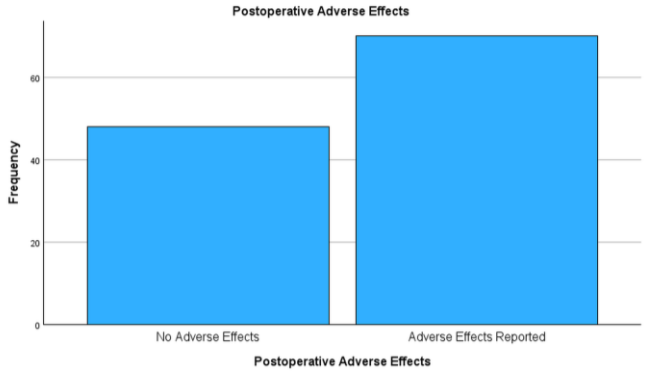


Figure 48: Frequency Distribution and Bar Chart of Postoperative Adverse Effects Among 118 Patients.

Among 118 patients, 70 (59.3%) reported postoperative adverse effects, while 48 (40.7%) reported no adverse effects. The bar chart shows that more than half of patients reported adverse effects, with a difference of 22 patients between the two categories.

8. Results and Interpretation

8.1 Descriptive Statistical Findings

Descriptive statistical analysis was performed on 118 patient records, including 59 patients who received IV Paracetamol and 59 who received IV Ketorolac. The analysis examined patient demographics, clinical characteristics, postoperative pain severity, satisfaction, pain relief, and adverse effects.

The largest age group was 18–25 years, comprising 49 patients (41.5%). The sample included 68 males (57.6%) and 50 females (42.4%). Most patients had a BMI of 18.5–24.9 (48.3%), followed by 25.0–29.9 (28.8%). ASA physical status I was the most common classification, recorded in 74 patients (62.7%).

Postoperative pain assessment showed that 41 patients (34.7%) experienced severe pain, 37 (31.4%) reported moderate pain, and 36 (30.5%) experienced mild pain. Only four patients (3.4%) reported the worst pain.

Patient satisfaction results showed that 40 patients (33.9%) were very satisfied and 37 (31.4%) were satisfied. Adequate postoperative pain relief was reported by 85 patients (72.0%), while 16 (13.6%) reported inadequate relief and 17 (14.4%) were uncertain.

Seventy patients (59.3%) reported postoperative adverse effects, while 48 (40.7%) reported no adverse effects.

These descriptive findings summarize patient characteristics and postoperative outcomes before comparing the two treatment groups.

8.2 Postoperative Pain Control Comparison

We compared postoperative pain severity between the IV Paracetamol and IV Ketorolac groups using the Visual Analogue Scale (VAS). Each treatment group included 59 patients.

In the IV Paracetamol group, 18 patients (30.5%) reported mild pain, 19 (32.2%) reported moderate pain, 20 (33.9%) experienced severe pain, and 2 (3.4%) reported worst pain. Similarly, in the IV Ketorolac group, 18 patients (30.5%) reported mild pain, 18 (30.5%) reported moderate pain, 21 (35.6%) experienced severe pain, and 2 (3.4%) reported worst pain.

The Pearson Chi-Square Test showed no statistically significant association between treatment group and postoperative VAS pain severity, $\chi^2(3, N = 118) = 0.051$, $p = 0.997$. Because two expected cell counts were below five, we also performed the Fisher-Freeman-Halton Exact Test, which confirmed a non-significant result ($p = 1.000$).

The Mann-Whitney U Test further supported these findings ($U = 1721.000$, $Z = -0.111$, $p = 0.912$). The mean ranks were 59.17 for IV Paracetamol and 59.83 for IV Ketorolac.

These findings indicate that postoperative VAS pain severity distributions were similar between the two treatment groups. Neither treatment demonstrated a statistically significant advantage in pain severity based on the recorded VAS categories.

8.3 Patient Satisfaction Comparison

Patient satisfaction was compared between the IV Paracetamol and IV Ketorolac groups, with 59 patients in each group. Satisfaction was assessed using five categories, ranging from very dissatisfied to very satisfied.

In the IV Paracetamol group, 31 patients (52.5%) were very satisfied, 16 (27.1%) were satisfied, 8 (13.6%) were neutral, 1 (1.7%) was dissatisfied, and 3 (5.1%) were very dissatisfied. In comparison, the IV Ketorolac group included 9 patients (15.3%) who were very satisfied, 21 (35.6%) who were satisfied, 20 (33.9%) who were neutral, 6 (10.2%) who were dissatisfied, and 3 (5.1%) who were very dissatisfied.

Positive satisfaction, combining satisfied and very satisfied responses, was reported by 47 patients (79.7%) receiving IV Paracetamol and 30 patients (50.8%) receiving IV Ketorolac.

The Pearson Chi-Square Test identified a statistically significant association between treatment group and patient satisfaction, $\chi^2(4, N = 118) = 21.490$, $p < 0.001$. Because four expected cell counts (40%) were below five, we performed the Fisher-Freeman-Halton Exact Test, which confirmed the significant association (exact $p < 0.001$).

These findings indicate that patients receiving IV Paracetamol reported higher satisfaction than those receiving IV Ketorolac. However, patient satisfaction is a subjective outcome and does not independently establish superior pain control.

8.4 Adequacy of Pain Relief Comparison

We compared the adequacy of postoperative pain relief between the IV Paracetamol and IV Ketorolac groups, with 59 patients in each group. Patients assessed whether their pain relief was adequate using three responses: Yes, No, and Uncertain.

In the IV Paracetamol group, 49 patients (83.1%) reported adequate pain relief, while 5 (8.5%) reported inadequate relief and 5 (8.5%) were uncertain. In comparison, 36 patients (61.0%) receiving IV Ketorolac reported adequate pain relief, 11 (18.6%) reported inadequate relief, and 12 (20.3%) were uncertain.

The Pearson Chi-Square Test showed a statistically significant association between treatment group and adequacy of postoperative pain relief, $\chi^2(2, N = 118) = 7.121$, $p = 0.028$. All expected cell counts were above five, satisfying the test's expected-frequency assumption.

A higher proportion of patients receiving IV Paracetamol reported adequate pain relief than those receiving IV Ketorolac, a difference of 22.1 percentage points. These findings suggest that IV Paracetamol was associated with better patient-reported adequacy of pain relief in the analysed records. However, the VAS pain severity comparison showed no statistically significant difference between the treatment groups.

8.5 Adverse Effects Comparison

We compared postoperative adverse effects between the IV Paracetamol and IV Ketorolac groups, with 59 patients in each group. The analysis examined the frequency of patients reporting adverse effects following postoperative pain management.

In the IV Paracetamol group, 38 patients (64.4%) reported adverse effects, while 21 (35.6%) reported none. In comparison, 32 patients (54.2%) receiving IV Ketorolac reported adverse effects, while 27 (45.8%) experienced none. The difference in reported adverse effects between the two groups was 10.2 percentage points.

The recorded adverse effects included nausea, vomiting, dizziness, gastrointestinal discomfort, allergic reactions, breathing difficulties, bleeding-related complications, and restlessness. Some patients reported multiple adverse effects.

The Pearson Chi-Square Test showed no statistically significant association between treatment group and postoperative adverse effects, $\chi^2(1, N = 118) = 1.264$, $p = 0.261$. All expected cell counts were above five, satisfying the expected-frequency assumption. Fisher's Exact Test also produced a non-significant result (two-sided $p = 0.349$).

Although adverse effects were reported more frequently in the IV Paracetamol group, the difference was not statistically significant. Therefore, the available results do not provide sufficient evidence to conclude that either treatment had a lower risk of postoperative adverse effects.

8.6 Interpretation of P-Values

We evaluated statistical significance at the 0.05 level. We considered a p-value below 0.05 statistically significant, while a p-value of 0.05 or above indicated insufficient evidence to reject the null hypothesis.

Postoperative VAS pain severity showed no statistically significant association with treatment group using Pearson's Chi-Square Test ($p = 0.997$). The Fisher-Freeman-Halton Exact Test ($p = 1.000$) and Mann-Whitney U Test ($p = 0.912$) supported this finding. These results indicate that the recorded pain severity distributions were similar between the IV Paracetamol and IV Ketorolac groups.

Patient satisfaction was significantly associated with treatment group ($p < 0.001$). Patients receiving IV Paracetamol reported higher satisfaction than those receiving IV Ketorolac. Adequacy of postoperative pain relief was also significantly associated with treatment group ($p = 0.028$), with a higher proportion of patients reporting adequate relief in the IV Paracetamol group.

Postoperative adverse effects showed no statistically significant association with treatment group ($p = 0.261$). Although the IV Paracetamol group reported adverse effects more frequently, the difference was not statistically significant.

These findings suggest that treatment groups differed in reported patient satisfaction and adequacy of pain relief, but no significant differences were detected in VAS pain severity or adverse effects. The p-values were not adjusted for multiple comparisons. Therefore, interpret the results cautiously, particularly the pain relief finding.

8.7 Statistical Results Summary Table

Table 2: Summary of Inferential Statistical Results Comparing IV Paracetamol and IV Ketorolac (N = 118).

Outcome	Statistical test	Test statistic	p-value	Result
VAS Pain Severity	Pearson Chi-Square	$\chi^2(3) = 0.051$	0.997	Not significant
VAS Pain Severity	Fisher-Freeman-Halton Exact	0.209	1.000	Not significant
VAS Pain Severity	Mann-Whitney U	U = 1721.000	0.912	Not significant
Patient Satisfaction	Pearson Chi-Square	$\chi^2(4) = 21.490$	< 0.001	Significant
Adequacy of Pain Relief	Pearson Chi-Square	$\chi^2(2) = 7.121$	0.028	Significant
Adverse Effects	Pearson Chi-Square	$\chi^2(1) = 1.264$	0.261	Not significant

The statistical analysis identified significant associations between treatment group and patient satisfaction ($p < 0.001$) and adequacy of postoperative pain relief ($p = 0.028$). Positive satisfaction was reported by 79.7% of patients receiving IV Paracetamol compared with 50.8% receiving IV Ketorolac. Similarly, 83.1% and 61.0% of patients reported adequate pain relief, respectively.

No statistically significant differences were detected in postoperative VAS pain severity or adverse effects. These results suggest that IV Paracetamol was associated with better patient-reported satisfaction and adequacy of pain relief in the analysed records. However, the findings do not establish that IV Paracetamol provides superior pain reduction or has a better safety profile.

The analysis included all 118 available patient records. Final interpretation remains subject to verification of patient eligibility and the study's inclusion and exclusion criteria.

9. Discussion

9.1 Interpretation of Major Findings

The study compared the effectiveness of IV Paracetamol and IV Ketorolac for postoperative pain management following elective hernia repair under spinal anaesthesia. The analysis included 118 patient records, with 59 patients in each treatment group. The main outcomes were postoperative pain severity, patient satisfaction, adequacy of pain relief, and adverse effects.

The results showed no statistically significant difference in postoperative Visual Analogue Scale (VAS) pain severity between the two treatment groups ($p = 0.997$). The Mann-Whitney U Test also showed a non-significant difference ($p = 0.912$). These findings indicate that the recorded pain severity distributions were similar for patients receiving IV Paracetamol and IV Ketorolac.

However, patient satisfaction differed significantly between the groups ($p < 0.001$). In the IV Paracetamol group, 79.7% of patients reported being satisfied or very satisfied, compared with 50.8% in the IV Ketorolac group. Adequate pain relief was also reported more frequently with IV Paracetamol (83.1%) than IV Ketorolac (61.0%), with a statistically significant association ($p = 0.028$).

The difference between VAS pain severity and patient-reported pain relief is important. Pain scores measure pain intensity, while satisfaction and perceived adequacy may also reflect patient expectations, comfort, and personal experiences. Therefore, higher satisfaction does not necessarily mean that one medication provides greater pain reduction.

Adverse effects were reported by 64.4% of patients receiving IV Paracetamol and 54.2% receiving IV Ketorolac. However, the difference was not statistically significant ($p = 0.261$). The findings do not establish that either medication has a better safety profile.

These results suggest that IV Paracetamol was associated with better patient-reported satisfaction and adequacy of pain relief in the analysed records, while measured pain severity and adverse-effect rates did not differ significantly.

9.2 Comparison with Previous Research

Previous research has examined the effectiveness of paracetamol and non-steroidal anti-inflammatory drugs (NSAIDs) in postoperative pain management.

Chaudhary et al. (2022) conducted a randomized controlled trial involving 80 patients undergoing gynecological abdominal surgery. The researchers compared IV Paracetamol with IV Ketorolac and found significantly lower mean postoperative pain scores in the Ketorolac group (3.05 ± 0.61) than in the Paracetamol group (3.56 ± 0.89), with $p = 0.04$.

This differs from the present analysis, where VAS pain severity did not differ significantly between treatment groups. This variation may reflect differences in surgical procedures, patient characteristics, anaesthetic methods, medication administration, and pain assessment. The previous trial meas-

ured pain using a numerical rating scale at 12 hours after surgery, whereas the present analysis used recorded VAS severity categories.

Mulita et al. (2020) investigated three postoperative analgesic regimens among 259 patients undergoing open inguinal hernia repair. The study compared IV Paracetamol alone with combinations of IV Paracetamol and either pethidine or parecoxib. Patients receiving combination treatment reported significantly lower postoperative pain scores than those receiving Paracetamol alone ($p < 0.0001$).

These findings support the possible benefits of combining appropriate analgesic medications. However, the study did not directly compare IV Paracetamol with IV Ketorolac, so its results cannot establish which of these two medications is more effective.

Ahmed et al. (2024) conducted a randomized controlled trial involving 60 patients undergoing open inguinal hernia mesh repair. The researchers found that IV Ketorolac provided better postoperative pain control than IV Tramadol, with earlier recovery-related improvements. Although this supports Ketorolac's effectiveness in hernia surgery, the comparison involved Tramadol rather than Paracetamol.

Coppens et al. (2020) reviewed evidence on postoperative pain management following open inguinal hernia repair. Their recommendations support using paracetamol alongside an NSAID or a COX-2-selective inhibitor, where clinically appropriate. The review also recommends suitable local or regional analgesic techniques to improve postoperative pain control.

The present findings are consistent with previous evidence showing that both Paracetamol and NSAIDs have roles in postoperative analgesia. However, differences in study designs and treatment regimens limit direct comparisons. The higher satisfaction observed with IV Paracetamol in the current records should be investigated further before concluding its clinical superiority.

9.3 Clinical Implications

The findings provide useful information for postoperative pain management following elective hernia repair under spinal anaesthesia.

First, both IV Paracetamol and IV Ketorolac were associated with similar recorded postoperative VAS pain severity distributions. This suggests that medication selection should consider the patient's clinical condition, contraindications, treatment needs, and response to analgesia rather than relying on a single outcome.

Second, the higher satisfaction and perceived adequacy of pain relief reported in the IV Paracetamol group show the importance of patient-reported outcomes. Healthcare professionals should assess pain intensity together with patient comfort and satisfaction.

Third, the relatively high frequency of reported adverse effects in both groups supports careful postoperative monitoring. Ketorolac requires particular attention to gastrointestinal bleeding and renal risks, while Paracetamol administration requires appropriate dosing and consideration of liver-related risks.

Fourth, postoperative pain management should be individualised. Coppens et al. (2020) recommend multimodal analgesia for open inguinal hernia repair, including appropriate non-opioid medications and regional techniques. Such approaches may provide better pain management than relying on a single medication.

Finally, regular pain assessment and documentation are essential. Standardised pain measurements and patient feedback can help healthcare professionals identify inadequate relief and adjust treatment to meet clinical needs.

9.4 Limitations of the Study

Several limitations should be considered when interpreting the results.

The analysis included 118 patient records from the available dataset. This sample may not provide sufficient statistical power to detect small differences between treatment groups, particularly for adverse effects.

The analysis compared treatment groups without adjusting for possible differences in patient characteristics, such as age, BMI, ASA physical status, and relevant medical history. These factors may influence postoperative pain, satisfaction, and medication safety.

We analysed pain severity using four VAS categories rather than continuous pain scores. Categorising pain measurements reduces detail and may make smaller differences between treatment groups difficult to identify.

We based patient satisfaction and adequacy of pain relief on subjective responses. Individual expectations, previous experiences, communication with healthcare professionals, and other aspects of postoperative care may influence these outcomes.

The present analysis did not independently verify complete information about medication doses, administration times, rescue analgesics, or consistent postoperative assessment times. These details are important when comparing analgesic effectiveness.

Adverse effects were analysed mainly as reported or not reported. This approach does not fully explain the severity, timing, or specific causes of individual reactions.

In addition, the study involved several statistical comparisons without adjustment for multiple testing. The significant finding for adequacy of pain relief ($p = 0.028$) should therefore be interpreted cautiously.

Patient eligibility and compliance with the original inclusion and exclusion criteria also require final verification before confirming the study sample.

These limitations mean that the findings should be interpreted as associations within the analysed records rather than definitive evidence of treatment superiority.

10. Conclusion and Recommendations

10.1 Conclusion

The study compared postoperative pain management outcomes between IV Paracetamol and IV Ketorolac following elective hernia repair under spinal anaesthesia. The study analysed 118 patient records, with 59 patients in each treatment group.

The results showed no statistically significant difference in postoperative VAS pain severity between the IV Paracetamol and IV Ketorolac groups ($p = 0.997$). The Mann-Whitney U Test supported this finding ($p = 0.912$).

However, patient satisfaction was significantly higher in the IV Paracetamol group ($p < 0.001$). Patients receiving IV Paracetamol also reported adequate pain relief more frequently (83.1%) than those receiving IV Ketorolac (61.0%), with $p = 0.028$.

Postoperative adverse effects were reported by 59.3% of all patients. Although the proportion was higher in the IV Paracetamol group, the difference between treatment groups was not statistically significant ($p = 0.261$).

The findings suggest that IV Paracetamol was associated with greater patient satisfaction and better perceived adequacy of pain relief. Nevertheless, neither treatment demonstrated a statistically significant advantage in recorded VAS pain severity or adverse effects.

Medication selection should therefore consider the patient's clinical condition, expected benefits, possible adverse reactions, and individual response to treatment. Further research is required to confirm whether the observed differences in patient-reported outcomes are clinically meaningful.

10.2 Recommendations

Based on the findings, the following recommendations are proposed:

1. Healthcare professionals should assess postoperative pain using standardised pain scales and record the results at regular intervals.
2. Patient satisfaction and perceived adequacy of pain relief should be assessed alongside pain intensity to provide a more complete understanding of treatment outcomes.
3. IV Paracetamol and IV Ketorolac should be selected according to individual patient needs, medical history, contraindications, and clinical assessment.
4. Patients receiving postoperative analgesics should be monitored for nausea, vomiting, dizziness, gastrointestinal symptoms, bleeding, and other relevant adverse reactions.
5. Healthcare professionals should consider evidence-based multimodal pain management when appropriate, rather than depending on one medication alone.
6. Accurate documentation of medication doses, administration times, rescue analgesics, and postoperative outcomes should be maintained.
7. Patients experiencing inadequate pain relief should receive timely reassessment and appropriate adjustment of their analgesic treatment.

These recommendations aim to support safe medication use, effective postoperative monitoring, and improved patient-centred care.

10.3 Suggestions for Future Research

Future research should include larger patient samples from multiple hospitals to improve the reliability and generalisability of the findings.

Prospective randomized controlled trials could provide stronger evidence by standardising medication doses, administration times, surgical procedures, and postoperative care.

Future studies should measure pain using continuous VAS or numerical rating scores at defined intervals, such as 2, 6, 12, 24, and 48 hours after surgery. This would help researchers examine changes in pain over time rather than relying on a single categorised result.

Researchers should also investigate additional outcomes, including time to first analgesic rescue, total analgesic consumption, mobilisation, length of hospital stay, and patient recovery.

Validated satisfaction questionnaires could improve the reliability of patient-reported outcomes. Adverse effects should be recorded separately by type, severity, timing, and possible relationship to treatment.

Further studies should examine whether factors such as age, BMI, ASA physical status, and existing medical conditions influence the response to IV Paracetamol and IV Ketorolac.

Future investigations could also compare individual medications with appropriate multimodal analgesic regimens to identify strategies that provide effective pain control with acceptable safety.

References

Ahmed, S. M., Shabbir, S., Rana, N. A., Khatoon, A., Ghani, U. F., Basharat, I., Khan, M. N., Hameed, F. M., & Dar, M. F. (2024). Comparing the postoperative analgesic efficacy of ketorolac and tramadol after open inguinal hernia mesh repair: A randomized controlled trial. *Cureus, 16*(10), Article e71363. https://doi.org/10.7759/cureus.71363

Chaudhary, S., Iqbal, S., Zia, R., Butt, M. M., Khan, M. B. A., & Abaid-Ur-Rehman. (2022). To compare mean postoperative pain scores between intravenous paracetamol and intravenous ketorolac among patients undergoing gynecological abdominal surgery. *Pakistan Journal of Medical & Health Sciences, 16*(9), 214. https://doi.org/10.53350/pjmhs22169214

Coppens, S., Gidts, J., Huynen, P., Van de Velde, M., & Joshi, G. P. (2020). Pain management after open inguinal hernia repair: An updated systematic review and procedure-specific postoperative pain management (PROSPECT/ESRA) recommendations. *Acta Anaesthesiologica Belgica, 71*, 45–56. https://esraeurope.org/prospect/procedures/inguinal-hernia-repair-2019/prospect-publication-open-inguinal-hernia-repair/

Mulita, F., Parchas, N., Solou, K., Tchabashvili, L., Gatomati, F., Iliopoulos, F., & Maroulis, I. (2020). Postoperative pain scores after open inguinal hernia repair: Comparison of three postoperative analgesic regimens. *Medical Archives, 74*(5), 355–358. https://doi.org/10.5455/medarh.2020.74.355-358